EG91 Series

Hardware Design

LTE Standard Module Series

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Safety Information

The following safety precautions must be observed during all phases of operation, such as usage, service or repair of any terminal or mobile incorporating the module. Manufacturers of the terminal should notify users and operating personnel of the following safety information by incorporating these guidelines into all manuals of the product. Otherwise, Quectel assumes no liability for customers' failure to comply with these precautions.

	Full attention must be paid to driving at all times in order to reduce the risk of an accident. Using a mobile while driving (even with a handsfree kit) causes distraction and can lead to an accident. Please comply with laws and regulations restricting the use of wireless devices while driving.
	Switch off the terminal or mobile before boarding an aircraft. The operation of wireless appliances in an aircraft is forbidden to prevent interference with communication systems. If there is an Airplane Mode, it should be enabled prior to boarding an aircraft. Please consult the airline staff for more restrictions on the use of wireless devices on an aircraft.
	Wireless devices may cause interference on sensitive medical equipment, so please be aware of the restrictions on the use of wireless devices when in hospitals, clinics or other healthcare facilities.
	Terminals or mobiles operating over radio signal and cellular network cannot be guaranteed to connect in certain conditions, such as when the mobile bill is unpaid or the (U)SIM card is invalid. When emergency help is needed in such conditions, use emergency call if the device supports it. In order to make or receive a call, the terminal or mobile must be switched on in a service area with adequate cellular signal strength. In an emergency, the device with emergency call function cannot be used as the only contact method considering network connection cannot be guaranteed under all circumstances.
	The terminal or mobile contains a transceiver. When it is ON, it receives and transmits radio frequency signals. RF interference can occur if it is used close to TV sets, radios, computers or other electric equipment.
	In locations with explosive or potentially explosive atmospheres, obey all posted signs and turn off wireless devices such as mobile phone or other terminals. Areas with explosive or potentially explosive atmospheres include fueling areas, below decks on boats, fuel or chemical transfer or storage facilities, and areas where the air contains chemicals or particles such as grain, dust or metal powders.

About the Document

Revision History

Version	Date	Author	Description
1.0	2017-03-22	Barret XIONG/ Rex WANG	Initial
1.1	2018-01-23	Barret XIONG/ Rex WANG	1. Added band B28A. 2. Updated the description of UMTS and GSM features in Table 2. 3. Updated the functional diagram in Figure 1. 4. Updated module operating frequencies in Table 21. 5. Updated current consumption in Table 26. 6. Updated RF output power in Table 27. 7. Updated the conducted RF receiving sensitivity in Table 28. 8. Updated the GPRS multi-slot classes in Table 33. 9. Added thermal consideration in Chapter 5.8 10. Added a GND pad in each of the four corners of the module's footprint in Chapter 6.2. 11. Updated storage information in Chapter 7.1. 12. Added packaging information in Chapter 7.3.
1.2	2018-03-14	Barret XIONG/ Rex WANG	1. Added the description of EG91-NA. 2. Updated the functional diagram in Figure 1. 3. Updated pin assignment in Figure 2. 4. Updated GNSS function in Table 1. 5. Updated GNSS Features in Table 2. 6. Updated reference circuit of USB interface in Figure 21. 7. Added description of GNSS receiver in Chapter 4. 8. Updated pin definition of RF antenna in Table

			21. 9. Updated module operating frequencies in Table 22. 10. Added description of GNSS antenna interface in Chapter 5.2. 11. Updated antenna requirements in Table 25. 12. Updated RF output power in Table 32.
1.3	2019-02-03	Barret XIONG/ Rex WANG	1. Added new variants EG91-NS, EG91-V, EG91-EC and related contents. 2. Opened pin 24 as ADC0 and added related contents. 3. Updated functional diagram (Figure 1) 4. Updated pin assignment (Figure 2) 5. Updated GNSS features (Table 2) 6. Added USB_BOOT interface information (Chapter 3.18) 7. Updated storage information (Chapter 8.1) 8. Updated module operating frequencies (Table 23) 9. Updated antenna requirements (Table 26) 10. Added current consumption of EG91-NS, EG91-V and EG91-EC (Table 32, 33 and 34) 11. Added conducted RF receiving sensitivity of EG91-NS, EG91-V and EG91-EC (Table 40, 41 and 42)
1.4	2019-03-29	Barret XIONG/ Rex WANG	1. Modified module name EG91-EC to EG91-EX, and EG91-V to EG91-VX 2. Added newly supported 9.x of Android USB serial driver (Table 2) 3. Modified the reflow temperature range as . 238ºC to 245ºC (Chapter 8.2)
1.5	2019-05-24	Barret XIONG/ Rex WANG	1. Updated comment of RF antenna (Table 4) 2. Updated EG91-EX current consumption (Table 36) 3. Updated EG91-EX conducted RF receiving sensitivity (Table 44)
1.6	2019-07-05	Barret XIONG/ Rex WANG	1. Updated supported protocols (Table 2). 2. Updated timing of turning on module (Figure 12). 3. DFOTA is developed. 4. Updated description of USB_BOOT interface and timing sequence for entering emergency download mode (Chapter 3.18 and Figure 29).

1.7	2019-08-09	Barret XIONG/ Rex WANG	1. Added ThreadX variant EG91-NAX and updated related contents (Table 1 and 4, Chapter 2.2, 2.3, 3.2 and 5). 2. Added related notes of SPI interface not supported on ThreadX modules (Chapter 3.1, 3.3 and 3.13). 3. Added current consumption of EG91-NAX (Table 37). 4. Updated EG91-NA conducted RF receiving sensitivity (Table 41). 5. Updated EG91-NS conducted RF receiving sensitivity (Table 42). 6. Added EG91-NAX conducted RF receiving sensitivity (Table 45).
1.8	2019-11-07	Barret XIONG/ Rex WANG	1. Removed related information of ThreadX OS. 2. Updated the supported USB serial drivers (Table 2) 3. Updated the notes for GNSS performance (Chapter 4.2). 4. Updated the AT command be used to disable the receive diversity (Chapter 5.1.3). 5. Updated the current consumption of LTE-FDD B13 (Table 43).
1.9	2020-10-21	Barret XIONG/ Rex WANG	1. Removed related information of EG91-NS module according to product promotion strategy. 2. Added EG91-AUX and related information (Table 1, 39 and 48). 3. Added EG91-NAXD and related information (Table 1, 37 and 47). 4. Updated pin assignment (Top View) (Figure 2) 5. Updated the AT command for the operation of ANT_MAIN and ANT_DIV (Chapter 5.1.3). 6. Added EG91-EX/-NA/-VX/-NAX/-NAXD/-AUX bottom dimensions (Figure 43). 7. Added the note about the standard that the package warpage level of the module conforms to (Chapter 7.1). 8. Updated top view of the module (Figure 45). 9. Added EG91-EX/-NA/-VX/-NAX/-NAXD/-AUX bottom view (Figure 47). 10. Updated module storage information (Chapter 8.1). 11. Updated the cooling down slope of reflow soldering thermal profile;

			Added a note to clarify the precautions if a conformal coating is necessary for the module (Chapter 8.2).
2.0	2021-07-01	Barret XIONG/ Rex WANG	1. Added the related information of EG91-NAL in the whole document. 2. Added the note of BOOT_CONFIG (Chapter 3.3&3.7.1)
2.1	2021-07-28	Barret XIONG/ Rex WANG	1. Updated supported USB serial drivers for Linux (Table 2). 2. Updated EG91-E bottom dimensions (top view) (Figure 42). 3. Updated EG91-AUX/-EX/-NA/-NAL/-NAX/ -NAXD/-VX bottom dimensions (top view) (Figure 43).
2.2	2022-05-07	Joe MA/ Frank WANG	1. Removed EG91-NAL. 2. Updated the module size information (Chapter 2.1 & Table 3 & Chapter 6.1). 3. Updated the USB serial drivers and the module weight (Table 3). 4. Updated EG91-VX power consumption (Table 46). 5. Updated the recommended footprint, and the top and bottom views of the module (Chapter 6.2 & 6.3). 6. Updated the max slope in the manufacturing and soldering (Chapter 7.2). 7. Updated the note description of label, conformal coating, ultrasonic cleaning and soldering (Chapter 7.2). 8. Updated the packaging specifications (Chapter 7.3).
2.3	2023-5-25	Thor LEI/ Woping WANG/ Gavin LU	1. Added an applicable module EG91-JP. 2. Updated pins 76 and77 of EG91-AUX from NC to GRFC9 and GRFC10 respectively. 3. Updated the USB serial drivers (Table 2). 4. Updated the pin assignment (Chapter 3.2). 5. Updated the voltage stability requirements (Chapter 3.6.2). 6. Updated the reference circuit of power supply (Figure 9). 7. Updated the operating frequency (Table 24). 8. Added the information about antenna tuner control interface (Chapter 4.1.2). 9. Updated the antenna design requirements

			(Table 38). 10. Updated the ramp-to-soak slop of soak zone, ramp-up slope and cool-down slope of reflow zone; Added related note (Chapter 7.2). 11. Added the mounting direction chapter (Chapter 7.3.3).
2.4	2024-05-18	Soley ZHANG/ Thor LEI/ Jayden WANG/ Woping WANG	1. Added the LTE modulations (Table 2). 2. Added a note on avoiding abnormal RF functions caused by current on the module's pins (Chapter 3.2). 3. Updated the description for turning on the module through PWRKEY (Chapter 3.7.1). 4. Added a note on hardware flow control design for UART interfaces (Chapter 3.12). 5. Added the operating frequency of LTE-FDD B28A (Table 23). 6. Updated the reference circuit for GNSS antenna (Figure 32). 7. Updated the top and side dimensions for EG91-NA/-NAX/-NAXD (Figure 44). 8. Updated the recommended thickness of stencil for the module and the reference document (Chapter 7.2 & Table 55). 9. Added the notes specifying that mercury-containing materials and environments containing corrosive gases should be avoided for module processing (Chapter 7.2).
2.5	2025-07-08	Soley ZHANG/ Lucius ZHANG	1. Added AGNSS feature description (Table 2). 2. Added the DC characteristics for PWRKEY pin (Table 4). 3. Updated reference circuit for RF antenna interface and added relevant notes (Chapter 4.1.5). 4. Added a note about the impedance condition for power consumption test (Chapter 5.4). 5. Updated the module coplanarity requirement (Chapter 6.1). 6. Updated the perspective for module bottom dimensions (Figures 45 & 46). 7. Updated the pre-baking time to 24 h (Chapter 7.1).

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1 Introduction

This document defines the EG91 series module and describes its air interface and hardware interfaces which are connected with your applications.

This document can help you quickly understand module interface specifications, electrical and mechanical details, as well as other related information of EG91 series. With application note and user guide, you can use EG91 series to design and set up mobile applications easily.

2 Product Overview

2.1. Frequency Bands and Functions

EG91 series is an embedded 4G wireless communication module with receive diversity. It supports LTE/WCDMA/GSM wireless communication, and provides data connectivity on LTE-FDD, DC-HSDPA, HSPA+, HSDPA, HSUPA, WCDMA, EDGE and GPRS networks. It can also provide voice functionality [1] to meet your specific application demands.

The following table shows the frequency bands and functions of the module.

Table 1: Frequency Bands and Functions

Module	LTE Band (with Rx-diversity)	WCDMA Band (with Rx-diversity)	GSM Band	GNSS [2]
EG91-AUX [3]	FDD: B1/B2/B3/B4/B5/B7/ B8/B28/B66	B1/B2/B5/B8	GSM850/ EGSM900/ DCS1800/ PCS1900	GPS, GLONASS, BDS, Galileo, QZSS
EG91-E	FDD: B1/B3/B7/B8/B20/ B28A	B1/B8	EGSM900/ DCS1800	-
EG91-EX	FDD: B1/B3/B7/B8/B20/B28	B1/B8	EGSM900/ DCS1800	GPS, GLONASS, BDS, Galileo, QZSS
EG91-JP	FDD: B1/B3/B8/B18/B19/ B26	-	-	
EG91-NA	FDD: B2/B4/B5/B12/B13	B2/B4/B5	-	
EG91-NAX	FDD: B2/B4/B5/B12/B13/ B25/B26	B2/B4/B5	-	
EG91-NAXD	FDD: B2/B4/B5/B12/B13/ B25/B26	B2/B4/B5	-	
EG91-VX	FDD: B4/B13	-	-	

[1] EG91 series contains **Data + Voice** version and **Data-only** version.
[2] GNSS function is optional.
[3] EG91-AUX does not support Rx-diversity.

EG91 series is an SMD type module which can be embedded into applications through its 106 LGA pins. With a compact profile of 29.0 mm × 25.0 mm × 2.3 mm (EG91-E), or 29.0 mm × 25.0 mm × 2.45 mm (EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX), it can meet almost all requirements for M2M applications.

The module is integrated with Internet service protocols like TCP, UDP and PPP. Extended AT commands have been developed for you to use these Internet service protocols easily.

2.2. Key Features

The following table describes the detailed features of the module.

Table 2: Key Features

Feature	Details
Power Supply	● Supply voltage range: 3.3–4.3 V ● Typical supply voltage: 3.8 V
Transmit Power	● Class 4 (33 dBm ±2 dB) for GSM850 ● Class 4 (33 dBm ±2 dB) for EGSM900 ● Class 1 (30 dBm ±2 dB) for DCS1800 ● Class 1 (30 dBm ±2 dB) for PCS1900 ● Class E2 (27 dBm ±3 dB) for GSM850 8-PSK ● Class E2 (27 dBm ±3 dB) for EGSM900 8-PSK ● Class E2 (26 dBm ±3 dB) for DCS1800 8-PSK ● Class E2 (26 dBm ±3 dB) for PCS1900 8-PSK ● Class 3 (23 dBm ±2 dB) for WCDMA bands ● Class 3 (23 dBm ±2 dB) for LTE-FDD bands
LTE Features	● Support up to non-CA Cat 1 FDD ● Support 1.4/3/5/10/15/20 MHz RF bandwidths ● DL modulations: QPSK,16QAM and 64QAM ● UL modulations: QPSK and 16QAM ● LTE-FDD: Max. 10 Mbps (DL); Max. 5 Mbps (UL)
UMTS Features	● Support 3GPP Rel-8 DC-HSDPA, HSPA+, HSDPA, HSUPA and WCDMA ● DL modulations: QPSK, 16QAM and 64QAM ● UL modulation: QPSK ● DC-HSDPA: Max. 42 Mbps (DL) ● HSUPA: Max. 5.76 Mbps (UL) ● WCDMA: Max. 384 kbps (DL); Max. 384 kbps (UL)
GSM Features	**R99:** ● CSD: 9.6 kbps

	GPRS: ● Support GPRS multi-slot class 33 (33 by default) ● Coding scheme: CS 1–4 ● Max. 107 kbps (DL); Max. 85.6 kbps (UL) **EDGE:** ● Support EDGE multi-slot class 33 (33 by default) ● Support GMSK and 8-PSK for different MCS (Modulation and Coding Scheme) ● Downlink coding schemes: MCS 1–9 ● Uplink coding schemes: MCS 1–9 ● Max. 296 kbps (DL); Max. 236.8 kbps (UL)
Internet Protocol Features	● Support TCP/UDP/PPP/FTP/FTPS/HTTP/HTTPS/NTP/PING/QMI/NITZ/MMS/SMTP/SSL/MQTT/FILE/CMUX/SMTPS protocols ● Support PAP and CHAP protocols for PPP connections
SMS	● Text and PDU modes ● Point-to-point MO and MT ● SMS cell broadcast ● SMS storage: ME by default
(U)SIM Interfaces	● Support 1.8 V and 3.0 V (U)SIM cards ● Support Dual SIM Single Standby
Audio Features	● Support one digital audio interface: PCM interface ● GSM: HR/FR/EFR/AMR/AMR-WB ● WCDMA: AMR/AMR-WB ● LTE: AMR/AMR-WB ● Support echo cancellation and noise suppression
PCM Interface	● Used for audio function with external codec ● Support 16-bit linear data format ● Support long frame synchronization and short frame synchronization ● Support master and slave modes, but must be the master in long frame synchronization
USB Interface	● Compliant with USB 2.0 specification (slave mode only); the data transmission rate can reach up to 480 Mbps ● Used for AT command communication, data transmission, GNSS NMEA sentence output, software debugging, firmware upgrade and voice over USB ● Support USB serial drivers for: Windows 10/11, Linux 2.6–6.7, Android 4.x–14.x, etc.
UART Interfaces	**Main UART:** ● Used for AT command communication and data transmission ● Baud rates reach up to 921600 bps, 115200 bps by default ● Support RTS and CTS hardware flow control **Debug UART:** ● Used for Linux console and log output

	• 115200 bps baud rate
SPI	• Provide a duplex, synchronous and serial communication link with the peripheral devices • Dedicated to one-to-one connection, without chip selection • 1.8 V operation voltage with clock rates up to 50 MHz
Rx-diversity	Support LTE/WCDMA Rx-diversity
GNSS Features (Optional)	• Protocol: NMEA 0183 • Data update rate: 1 Hz by default • Support AGNSS. For more details, see ***document [1]***
AT Commands	• Compliant with *3GPP TS 27.007* and *3GPP TS 27.005* • Quectel enhanced AT command
Network Indication	NETLIGHT pin for network activity status indication
Antenna Interfaces	• Main antenna interface (ANT_MAIN) • Rx-diversity antenna interface (ANT_DIV) [4] • GNSS antenna interface (ANT_GNSS) [5]
Physical Characteristics	• Size: (29.0 ±0.15) mm × (25.0 ±0.15) mm × (2.3 ±0.2) mm (EG91-E) (29.0 ±0.15) mm × (25.0 ±0.15) mm × (2.45 ±0.2) mm (EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX) • Package: LGA • Weight: approx. 4.1 g (EG91-E) approx. 4.2 g (EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX)
Temperature Ranges	• Normal operating temperature [6]: -35 °C to +75 °C • Extended operating temperature [7]: -40 °C to +85 °C • Storage temperature: -40 °C to +90 °C
Firmware Upgrade	• USB interface • DFOTA
RoHS	All hardware components are fully compliant with EU RoHS directive

[4] Rx-diversity antenna is only supported on EG91-E/-EX/-JP/-NA/-NAX/-NAXD/-VX.
[5] GNSS antenna is only supported on EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX and the GNSS function is optional.
[6] To meet the normal operating temperature range requirements, it is necessary to ensure effective thermal dissipation, e.g., by adding passive or active heatsinks, heat pipes, vapor chambers. Within this range, the module's indicators comply with 3GPP specification requirements.
[7] To meet the extended operating temperature range requirements, it is necessary to ensure effective thermal dissipation, e.g., by adding passive or active heatsinks, heat pipes, vapor chambers. Within this range, the module retains the ability to establish and maintain functions such as voice, SMS and data transmission, without any unrecoverable malfunction. Radio spectrum and radio network remain uninfluenced, whereas the value of one or more parameters, such as P_{out}, may decrease and fall below the range of the 3GPP specified tolerances. When the temperature returns to the normal operating temperature range, the module's indicators will comply with 3GPP specification requirements again.

2.3. Functional Diagram

The following figure shows a block diagram of the module and illustrates the major functional parts.

- Power management
- Baseband
- DDR + NAND flash
- Radio frequency
- Peripheral interfaces

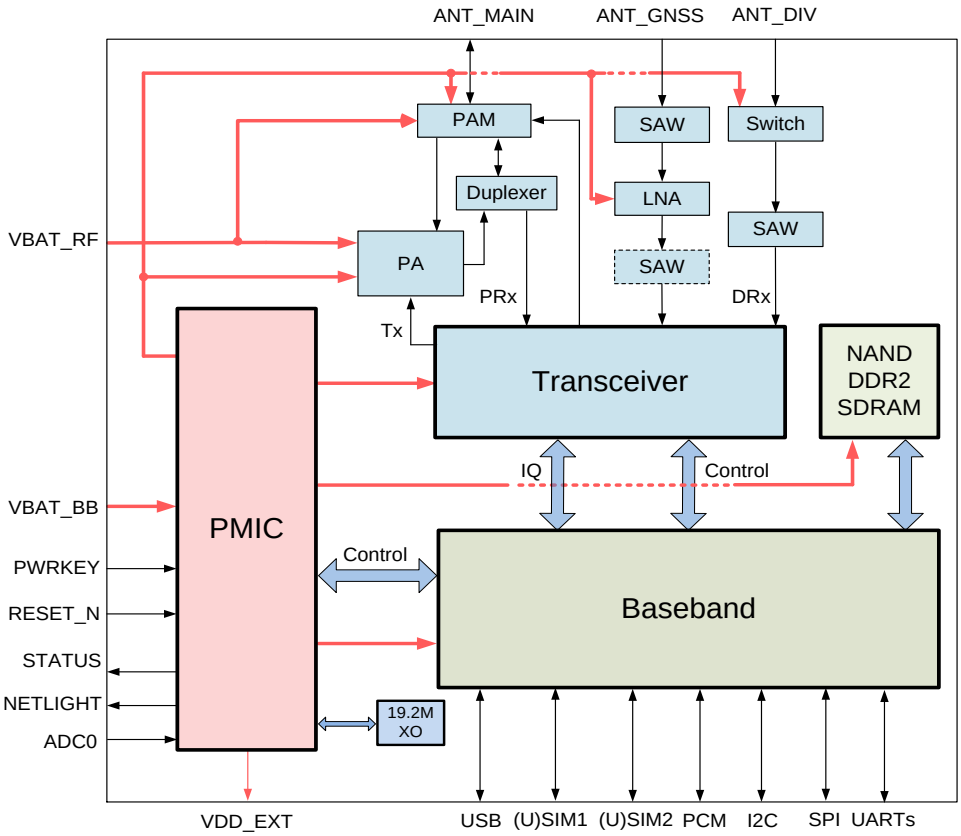


Figure 1: Functional Diagram [8]

2.4. EVB Kit

Quectel supplies an evaluation board (UMTS&LTE EVB) with accessories to develop and test the module. For more details, see ***document [2]***.

[8] GNSS antenna is only supported on EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX and the GNSS function is optional. Rx-diversity antenna is only supported on EG91-E/-EX/-JP/-NA/-NAX/-NAXD/-VX, and is not supported on EG91-AUX.

3 Application Interfaces

3.1. General Description

The module is equipped with 106 LGA pins that can be connected to your cellular application platforms. The subsequent chapters will provide detailed descriptions of the following interfaces/functions.

- Power supply
- (U)SIM interfaces
- USB interface
- UART interfaces
- PCM and I2C interfaces
- SPI
- Indication signals
- ADC interface
- USB_BOOT interface

3.2. Pin Assignment

The following figure shows the pin assignment of the module.

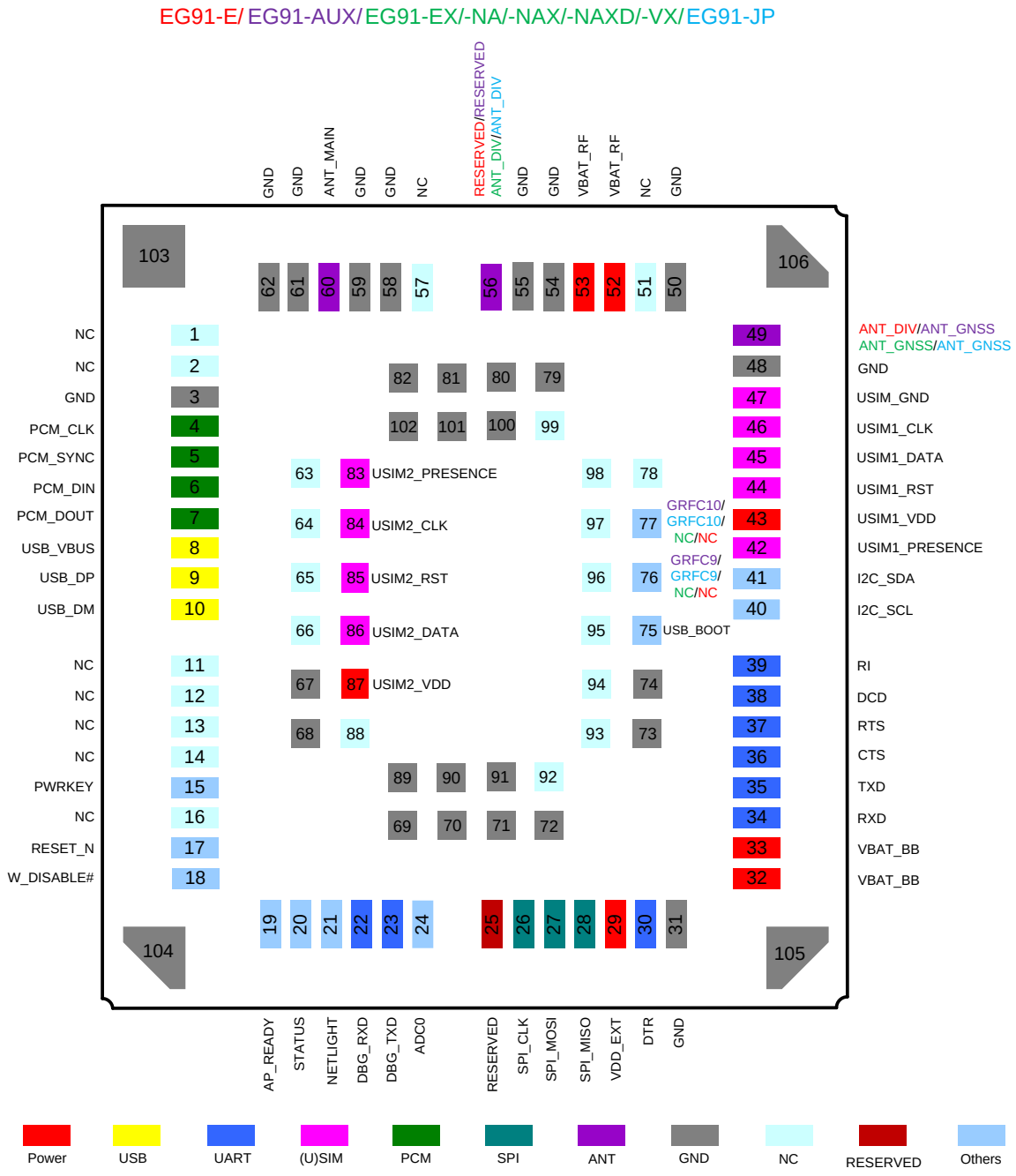


Figure 2: Pin Assignment (Top View)

NOTE

1. Ensure that the pull-up power supply of the module's pins is VDD_EXT or controlled by VDD_EXT, and there is no current sink on the module's pins before the module turns on. For more details, contact Quectel Technical Support.
2. PWRKEY output voltage is 0.8 V because of the diode drop in the baseband chipset.
3. Keep all RESERVED and unused pins unconnected.
4. GND pins should be connected to ground in the design.
5. Pin 49 is defined as ANT_GNSS on EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX, while it is defined as ANT_DIV on EG91-E.
6. Pin 56 is defined as ANT_DIV on EG91-EX/-JP/-NA/-NAX/-NAXD/-VX, while it is defined as RESERVED on EG91-AUX/-E. Rx-diversity antenna is not supported on EG91-AUX.
7. Pin 76 is defined as GRFC9 on EG91-AUX/-JP, while it is defined as NC on EG91-E/-EX/-NA/-NAX/-NAXD/-VX.
8. Pin 77 is defined as GRFC10 on EG91-AUX/-JP, while it is defined as NC on EG91-E/-EX/-NA/-NAX/-NAXD/-VX.

3.3. Pin Description

The following tables show the pin definition and description of the module.

Table 3: Parameter Definition

Parameter	Description
AI	Analog Input
AIO	Analog Input/Output
DI	Digital Input
DO	Digital Output
DIO	Digital Input/Output
OD	Open Drain
PI	Power Input
PO	Power Output

DC characteristics include power domain and rated current.

Table 4: Pin Description

Power Supply					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**
VBAT_BB	32, 33	PI	Power supply for the module's BB part	Vmax = 4.3 V Vmin = 3.3 V Vnom = 3.8 V	It must be provided with sufficient current up to 0.8 A. A test point is recommended to be reserved.
VBAT_RF	52, 53	PI	Power supply for the module's RF part		It must be provided with sufficient current up to 1.8 A in a burst transmission. A test point is recommended to be reserved.
VDD_EXT	29	PO	Provide 1.8 V for external circuit	Vnom = 1.8 V $I_O$max = 50 mA	Power supply for external GPIO's pull up circuits. If unused, keep it open.
GND	3, 31, 48, 50, 54, 55, 58, 59, 61, 62, 67–74, 79–82, 89–91, 100–106				
Turn-on/off					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**
PWRKEY	15	DI	Turn on/off the module	V_{IH} = 0.8 V V_{IL}max = 0.5 V	The output voltage is 0.8 V because of the diode drop in the baseband chipset. A test point is recommended to be reserved.
RESET_N	17	DI	Reset the module	V_{IH}max = 2.1 V V_{IH}min = 1.3 V V_{IL}max = 0.5 V	Require pull-up resistor to 1.8 V internally. Active LOW. If unused, keep it open.
Status Indication					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**

STATUS	20	DO	Indicate the module's operating status	V_{OH}min = 1.35 V V_{OL}max = 0.45 V	1.8 V power domain. If unused, keep them open.
NETLIGHT	21	DO	Indicate the module's network activity status		
USB Interface					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**
USB_VBUS	8	AI	USB connection detect	Vmax = 5.25 V Vmin = 3.0 V Vnom = 5.0 V	Typical value: 5.0 V If unused, keep it open.
USB_DP	9	AIO	USB differential data (+)		USB 2.0 compliant. Require differential impedance of 90 Ω.
USB_DM	10	AIO	USB differential data (-)		
(U)SIM Interfaces					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**
USIM_GND	47	-	Specified ground for (U)SIM card		Connect to ground of (U)SIM card connector.
USIM1_VDD	43	PO	(U)SIM1 card power supply	$I_O$max = 50 mA **For 1.8 V (U)SIM:** Vmax = 1.9 V Vmin = 1.7 V **For 3.0 V (U)SIM:** Vmax = 3.05 V Vmin = 2.7 V	Either 1.8 V or 3.0 V is supported by the module automatically.
USIM1_DATA	45	DIO	(U)SIM1 card data	**For 1.8 V (U)SIM:** V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{OL}max = 0.45 V V_{OH}min = 1.35 V **For 3.0 V (U)SIM:** V_{IL}max = 1.0 V V_{IH}min = 1.95 V V_{OL}max = 0.45 V V_{OH}min = 2.55 V	

USIM1_CLK	46	DO	(U)SIM1 card clock	**For 1.8 V (U)SIM:** V_{OL}max = 0.45 V V_{OH}min = 1.35 V **For 3.0 V (U)SIM:** V_{OL}max = 0.45 V V_{OH}min = 2.55 V	
USIM1_RST	44	DO	(U)SIM1 card reset		
USIM1_PRESENCE	42	DI	(U)SIM1 card hot-plug detect	V_{IL}min = -0.3 V V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{IH}max = 2.0 V	1.8 V power domain. If unused, keep it open.
USIM2_VDD	87	PO	(U)SIM2 card power supply	$I_O$max = 50 mA **For 1.8 V (U)SIM:** Vmax = 1.9 V Vmin = 1.7 V **For 3.0 V (U)SIM:** Vmax = 3.05 V Vmin = 2.7 V	Either 1.8 V or 3.0 V is supported by the module automatically. If unused, keep it open.
USIM2_DATA	86	DIO	(U)SIM2 card data	**For 1.8 V (U)SIM:** V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{OL}max = 0.45 V V_{OH}min = 1.35 V **For 3.0 V (U)SIM:** V_{IL}max = 1.0 V V_{IH}min = 1.95 V V_{OL}max = 0.45 V V_{OH}min = 2.55 V	If unused, keep them open.
USIM2_CLK	84	DO	(U)SIM2 card clock	**For 1.8 V (U)SIM:** V_{OL}max = 0.45 V V_{OH}min = 1.35 V **For 3.0 V (U)SIM:** V_{OL}max = 0.45 V V_{OH}min = 2.55 V	
USIM2_RST	85	DO	(U)SIM2 card reset	**For 1.8 V (U)SIM:** V_{OL}max = 0.45 V V_{OH}min = 1.35 V **For 3.0 V (U)SIM:** V_{OL}max = 0.45 V	

				V_{OH}min = 2.55 V	
USIM2_ PRESENCE	83	DI	(U)SIM2 card hot-plug detect	V_{IL}min = -0.3 V V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{IH}max = 2.0 V	1.8 V power domain. If unused, keep it open.
Main UART Interface					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**
RI	39	DO	Ring indication	V_{OL}max = 0.45 V V_{OH}min = 1.35 V	1.8 V power domain. If unused, keep them open.
DCD	38	DO	Data carrier detect		
CTS	36	DO	Clear to send signal from the module		Connect to MCU's CTS. 1.8 V power domain. If unused, keep it open.
RTS	37	DI	Request to send signal to the module	V_{IL}min = -0.3 V V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{IH}max = 2.0 V	Connect to MCU's RTS. 1.8 V power domain. If unused, keep it open.
DTR	30	DI	Data terminal ready; Sleep mode control		1.8 V power domain. Pulled up by default. Low level wakes up the module. If unused, keep it open.
TXD	35	DO	Transmit	V_{OL}max = 0.45 V V_{OH}min = 1.35 V	1.8 V power domain. If unused, keep them open.
RXD	34	DI	Receive	V_{IL}min = -0.3 V V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{IH}max = 2.0 V	
Debug UART Interface					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**
DBG_TXD	23	DO	Debug UART transmit	V_{OL}max = 0.45 V V_{OH}min = 1.35 V	1.8 V power domain. If unused, keep them open.
DBG_RXD	22	DI	Debug UART receive	V_{IL}min = -0.3 V V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{IH}max = 2.0 V	
PCM Interface					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**

PCM_DIN	6	DI	PCM data input	V_{IL}min = -0.3 V V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{IH}max = 2.0 V	1.8 V power domain. If unused, keep them open.
PCM_DOUT	7	DO	PCM data output	V_{OL}max = 0.45 V V_{OH}min = 1.35 V	
PCM_SYNC	5	DIO	PCM data frame sync	V_{OL}max = 0.45 V V_{OH}min = 1.35 V V_{IL}min = -0.3 V V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{IH}max = 2.0 V	1.8 V power domain. In master mode, they are output signals. In slave mode, they are input signals. If unused, keep them open.
PCM_CLK	4	DIO	PCM clock		
I2C Interface					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**
I2C_SCL	40	OD	I2C serial clock (for external codec)		Require an external pull-up to 1.8 V If unused, keep them open.
I2C_SDA	41	OD	I2C serial data (for external codec)		
ADC Interface					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**
ADC0	24	AI	General-purpose ADC interface	Input voltage range: 0.3 V to VBAT_BB	If unused, keep it open.
SPI					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**
SPI_CLK	26	DO	SPI clock	V_{OL}max = 0.45 V V_{OH}min = 1.35 V	1.8 V power domain. If unused, keep them open.
SPI_MOSI	27	DO	SPI master-out slave-in		
SPI_MISO	28	DI	SPI master-in slave-out	V_{IL}min = -0.3 V V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{IH}max = 2.0 V	
RF Interfaces					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**

ANT_GNSS/ ANT_DIV [9]	49	AI	GNSS antenna interface/Diversity antenna interface		50 Ω characteristic impedance. If unused, keep them open.
ANT_DIV/ RESERVED [10]	56	AI/ -	Diversity antenna interface/Reserved		
ANT_MAIN	60	AIO	Main antenna Interface		50 Ω characteristic impedance.
Other Pins					
Pin Name	**Pin No.**	**I/O**	**Description**	**DC Characteristics**	**Comment**
W_DISABLE#	18	DI	Airplane mode control	V_{IL}min = -0.3 V V_{IL}max = 0.6 V V_{IH}min = 1.2 V V_{IH}max = 2.0 V	1.8 V power domain. Pull-up by default. At low voltage level, the module can enter airplane mode. If unused, keep it open.
AP_READY	19	DI	Application processor ready		1.8 V power domain. If unused, keep it open.
USB_BOOT	75	DI	Force the module to enter download mode		1.8 V power domain. Active HIGH. A test point is recommended to be reserved.
GRFC9/NC [11]	76	DO/-	Generic RF controller/ Not connected		If unused, keep them open.
GRFC10/NC [11]	77	DO/-	Generic RF controller/ Not connected		
NC and RESERVED Pins					
Pin Name	**Pin No.**				**Comment**
NC [11]	1, 2, 11–14, 16, 51, 57, 63–66, 76–78, 88, 92–99				Keep them unconnected.
RESERVED [10]	25, 56				

[9] Pin 49 is defined as ANT_GNSS on EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX, while it is defined as ANT_DIV on EG91-E.
[10] Pin 56 is defined as ANT_DIV on EG91-EX/-JP/-NA/-NAX/-NAXD/-VX, while it is defined as RESERVED on EG91-AUX/-E. Rx-diversity antenna is not supported on EG91-AUX.
[11] Pin 76 is defined as GRFC9 on EG91-AUX/-JP, while it is defined as NC on EG91-E/-EX/-NA/-NAX/-NAXD/-VX. Pin 77 is defined as GRFC10 on EG91-AUX/-JP, while it is defined as NC on EG91-E/-EX/-NA/-NAX/-NAXD/-VX.

NOTE

1. Keep all RESERVED and unused pins unconnected.
2. USB_BOOT pin and BOOT_CONFIG pins (SPI_CLK, PCM_CLK, PCM_SYNC) cannot be pulled up before startup.

3.4. Operating Modes

The table below briefly outlines the operating modes to be mentioned in the following chapters.

Table 5: Overview of Operating Modes

Mode	Details	
Full Functionality Mode	Idle	Software is active. The module has registered on network, and it is ready to send and receive data.
	Voice/Data	Network is connected. In this mode, the power consumption is decided by network setting and data transmission rate.
Airplane Mode	**AT+CFUN=4** or pulling down W_DISABLE# pin can set the module to enter airplane mode. In this case, RF function will be invalid.	
Minimum Functionality Mode	**AT+CFUN=0** can set the module to a minimum functionality mode without removing the power supply. In this case, both RF function and (U)SIM card will be invalid.	
Sleep Mode	In this mode, the power consumption of the module will be reduced to an ultra-low level. During this mode, the module can still receive paging message, SMS, voice call and TCP/UDP data from the network normally.	
Turn-off Mode	In this mode, the power management unit shuts down the power supply. Software goes inactive. The UART interface is not accessible. Operating voltage (connected to VBAT_RF and VBAT_BB) remains applied.	

For details of the above AT command, see ***document [3]***.

3.5. Power Saving

3.5.1. Sleep Mode

The module is able to reduce its power consumption to an ultra-low level during the sleep mode. The following sub-chapters describe the power saving procedures of the module.

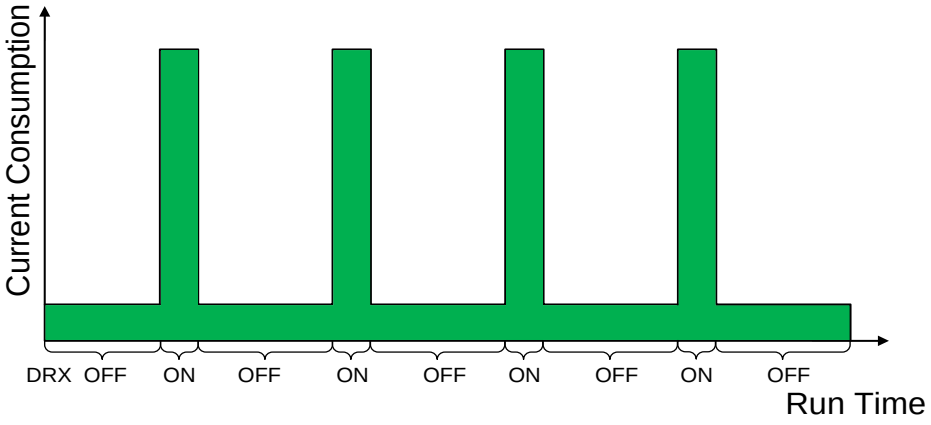


Figure 3: Power Consumption in Sleep Mode

NOTE

DRX cycle values are transmitted over the wireless network.

3.5.1.1. UART Application Scenario

If the MCU communicates with the module via UART interface, the following preconditions can let the module enter sleep mode.

- Execute **AT+QSCLK=1** to enable sleep mode.
- Drive DTR to high level.

The following figure shows the connection between the module and the MCU.

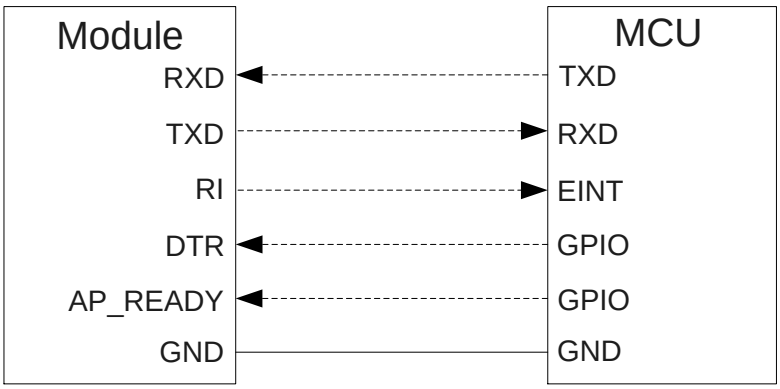


Figure 4: Sleep Mode Application via UART

- Driving the module's DTR to low level will wake up the module.
- When the module has a URC to report, RI signal will wake up the MCU. See ***Chapter 3.15.3*** for details about RI behaviors.
- AP_READY will detect the sleep state of MCU (can be configured to high-level or low-level detection). See **AT+QCFG="apready"** in ***document [4]*** for details.

3.5.1.2. USB Application with USB Remote Wakeup Function

If the host supports USB Suspend/Resume and remote wakeup functions, the following three preconditions must be met to let the module enter sleep mode.

- Execute **AT+QSCLK=1** to enable sleep mode.
- Ensure the DTR is held at high level or keep it open.
- The host's USB bus, which is connected with the module's USB interface, enters Suspend state.

The following figure shows the connection between the module and the host.

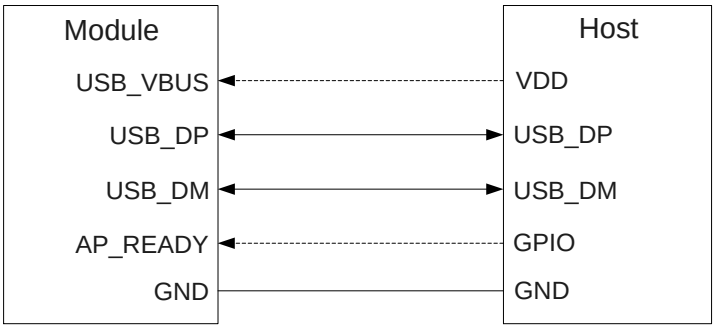


Figure 5: Sleep Mode Application with USB Remote Wakeup

- Sending data to the module through USB will wake it up.
- When the module has a URC to report, it will send remote wakeup signals via USB bus to wake up the host.

3.5.1.3. USB Application with USB Suspend/Resume and RI Function

If the host supports USB Suspend/Resume, but does not support remote wakeup function, the RI signal is needed to wake up the host.

There are three preconditions to let the module enter sleep mode.

- Execute **AT+QSCLK=1** to enable sleep mode.
- Ensure the DTR is held at high level or keep it open.
- The host's USB bus, which is connected with the module's USB interface, enters Suspend state.

The following figure shows the connection between the module and the host.

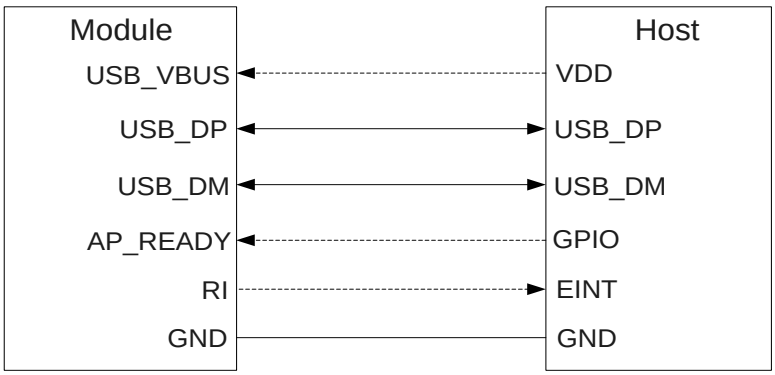


Figure 6: Sleep Mode Application with RI

- Sending data to the module through USB will wake it up.
- When the module has a URC to report, RI signal will wake up the host.

3.5.1.4. USB Application Without USB Suspend Function

If the host does not support USB Suspend function, USB_VBUS should be disconnected with an external control circuit to let the module enter sleep mode.

- Execute **AT+QSCLK=1** to enable sleep mode.
- Ensure the DTR is held at high level or keep it open.
- Disconnect USB_VBUS.

The following figure shows the connection between the module and the host.

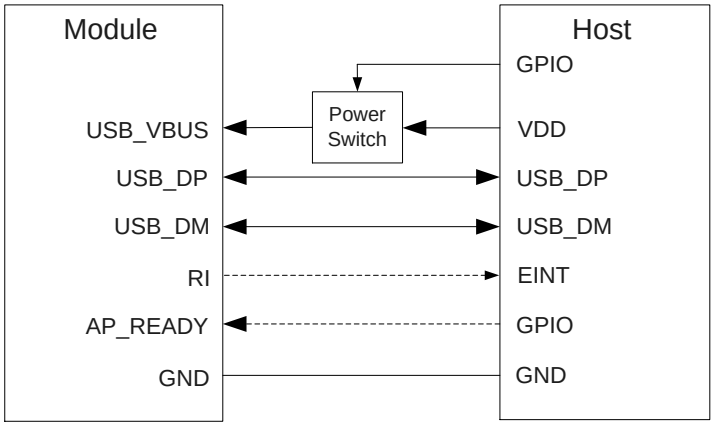


Figure 7: Sleep Mode Application Without Suspend Function

Restoring the power supply to USB_VBUS will wake up the module.

NOTE

1. Pay attention to the level match shown in dotted line between the module and the host.
2. For more details about the module power management application, see ***document [5]***.
3. For details of **AT+QSCLK**, see ***document [3]***.
4. For details of **AT+QCFG**, see ***document [4]***.

3.5.2. Airplane Mode

When the module enters airplane mode, the RF function will be disabled, and all AT commands related to it will be inaccessible. This mode can be set via the following ways.

Hardware:

The W_DISABLE# pin is pulled up by default. Driving it to low level will let the module enter airplane mode.

Software:

AT+CFUN=<fun> provides the choice of the functionality level through setting **<fun>** as 0, 1 or 4.

- **AT+CFUN=0:** Minimum functionality mode. Both (U)SIM and RF functions are disabled.
- **AT+CFUN=1:** Full functionality mode (by default).
- **AT+CFUN=4:** Airplane mode (RF function is disabled).

NOTE

1. Airplane mode control via W_DISABLE# is disabled in firmware by default. It can be enabled by **AT+QCFG="airplanecontrol"**. See ***document [4]*** for more details.
2. The execution of **AT+CFUN** will not affect GNSS function.

3.6. Power Supply

3.6.1. Power Supply Pins

The module provides four VBAT pins for connection with an external power supply. There are two separate voltage domains for VBAT.

- Two VBAT_RF pins for module's RF part.
- Two VBAT_BB pins for module's BB part.

The following table shows the details of VBAT and ground pins.

Table 6: Pin Definition of VBAT and GND

Pin Name	Pin No.	Description	Min.	Typ.	Max.	Unit
VBAT_RF	52, 53	Power supply for the module's RF part	3.3	3.8	4.3	V
VBAT_BB	32, 33	Power supply for the module's BB part	3.3	3.8	4.3	V
GND	3, 31, 48, 50, 54, 55, 58, 59, 61, 62, 67–74, 79–82, 89–91, 100–106					

3.6.2. Voltage Stability Requirements

The power supply range of the module is from 3.3 V to 4.3 V. Please make sure that the input voltage will never drop below 3.3 V. The following figure shows the voltage drop during burst transmission in 2G network. The voltage drop will be less in 3G and 4G networks.

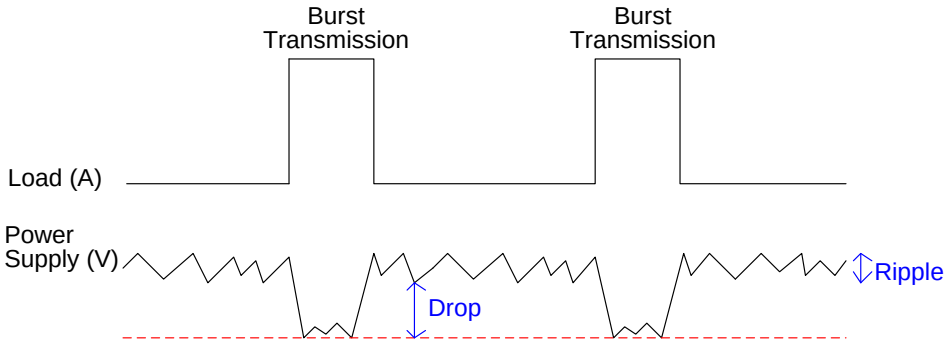


Figure 8: Power Supply Limits during Burst Transmission

To decrease voltage drop, a filter capacitor of about 100 µF with low ESR (ESR = 0.7 Ω) should be used for VBAT_BB and VBAT_RF respectively, and a multi-layer ceramic chip (MLCC) capacitor array should also be reserved due to its ultra-low ESR. It is recommended to use three ceramic capacitors (100 nF, 33 pF, 10/100 pF [12] for VBAT_BB and 100 nF, 33 pF, 10 pF for VBAT_RF) for composing the MLCC array, and place these capacitors close to VBAT_BB/VBAT_RF pins. The main power supply from an external application has to be a single voltage source and can be expanded to two sub paths with star structure.

The width of VBAT_BB trace should be at least 1 mm, and the width of VBAT_RF trace should be at least 2 mm. In principle, the longer the VBAT trace is, the wider it should be.

In addition, to avoid the damage caused by electric surge and ESD, it is suggested that a TVS diode with low reverse stand-off voltage V_{RWM} (4.5 V), low clamping voltage V_C and high reverse peak pulse current I_{PP} should be used. The following figure shows the star structure of the power supply.

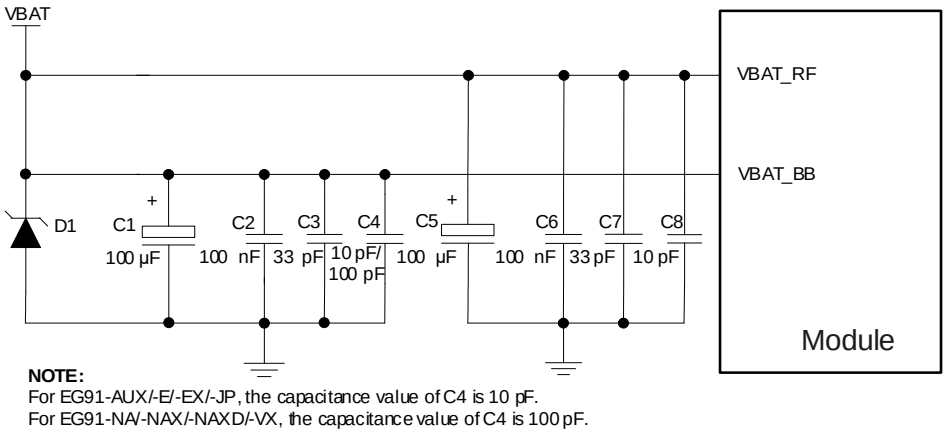


Figure 9: Star Structure of Power Supply

[12] For EG91-AUX/-E/-EX/-JP, the capacitance value is 10 pF. For EG91-NA/-NAX/-NAXD/-VX, the capacitance value is 100 pF.

3.6.3. Reference Design for Power Supply

Power design for the module is very important, as the performance of the module largely depends on the power source. The power supply should be able to provide sufficient current of at least 2 A. If the voltage drop between the input and output is not too high, it is suggested that an LDO should be used for the module. If there is a big voltage difference between the input source and the desired output (VBAT), a buck converter is preferred to be used as the power supply.

The following figure shows a reference design for +5 V input power source. The typical output of the power supply is about 3.8 V and the maximum load current is 3 A.

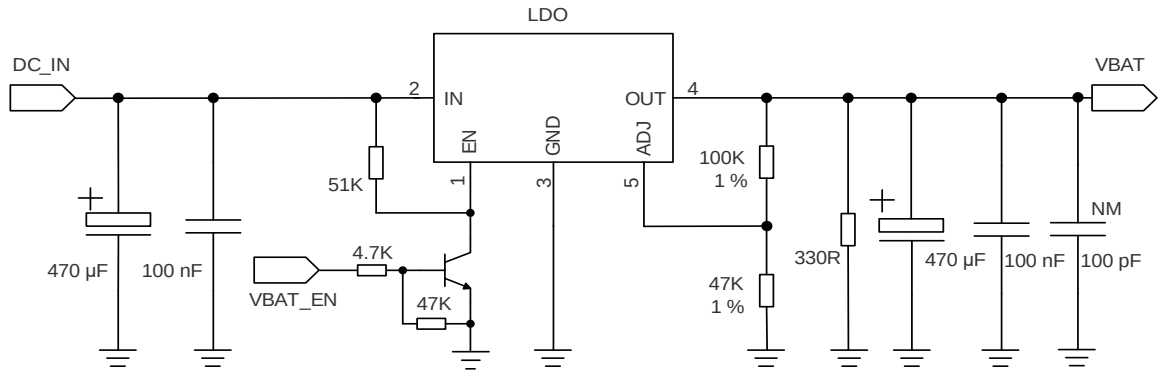


Figure 10: Reference Circuit of Power Supply

NOTE

To avoid corrupting the data in the internal flash, do not switch off the power supply when the module works normally. Only after the module is turned off by PWRKEY or AT command, then the power supply can be cut off.

3.6.4. Power Supply Voltage Monitoring

AT+CBC can be used to monitor the VBAT_BB voltage value. For more details, see ***document [3]***.

3.7. Turn-on

3.7.1. Turn-on with PWRKEY

The following table shows the pin definition of PWRKEY.

Table 7: Pin Definition of PWRKEY

Pin Name	Pin No.	Description	Comment
PWRKEY	15	Turn on/off the module	The output voltage is 0.8 V because of the diode drop in the baseband chipset. A test point is recommended to be reserved.

When the module is in turn-off mode, it can be turned on by driving the PWRKEY low for at least 500 ms. It is recommended to use an open drain/collector driver to control the PWRKEY, and release it after the STATUS pin outputs a high level. The STATUS pin is used as an indicator to show that the module has been turned on normally.

A simple reference circuit is illustrated in the following figure.

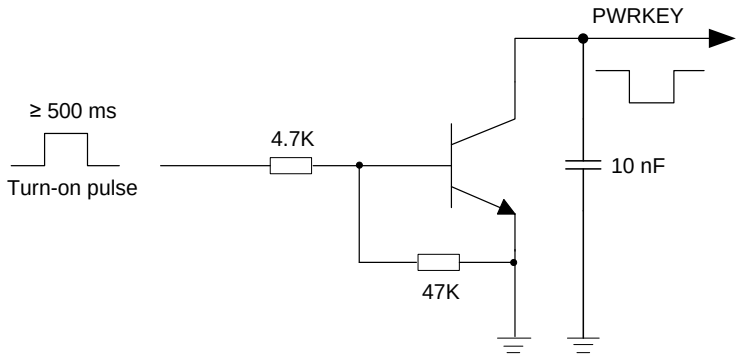


Figure 11: Turn On the Module Using Driving Circuit

Another way to control the PWRKEY is using a button directly. When pressing the button, electrostatic strike may be generated from the finger. Therefore, a TVS component is indispensable to be placed nearby the button for ESD protection. A reference circuit is shown in the following figure.

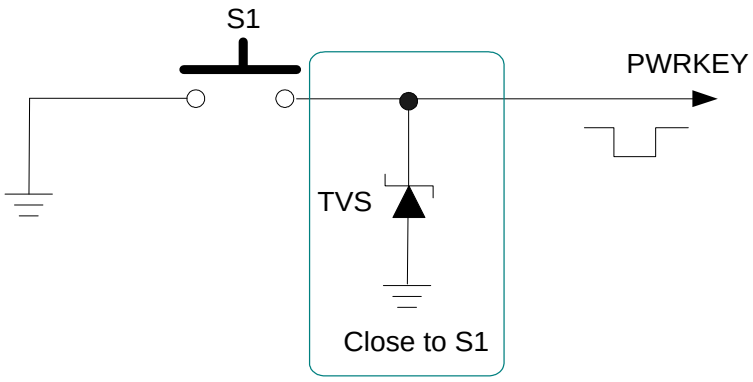


Figure 12: Turn On the Module Using a Button

The turn-on timing is illustrated in the following figure.

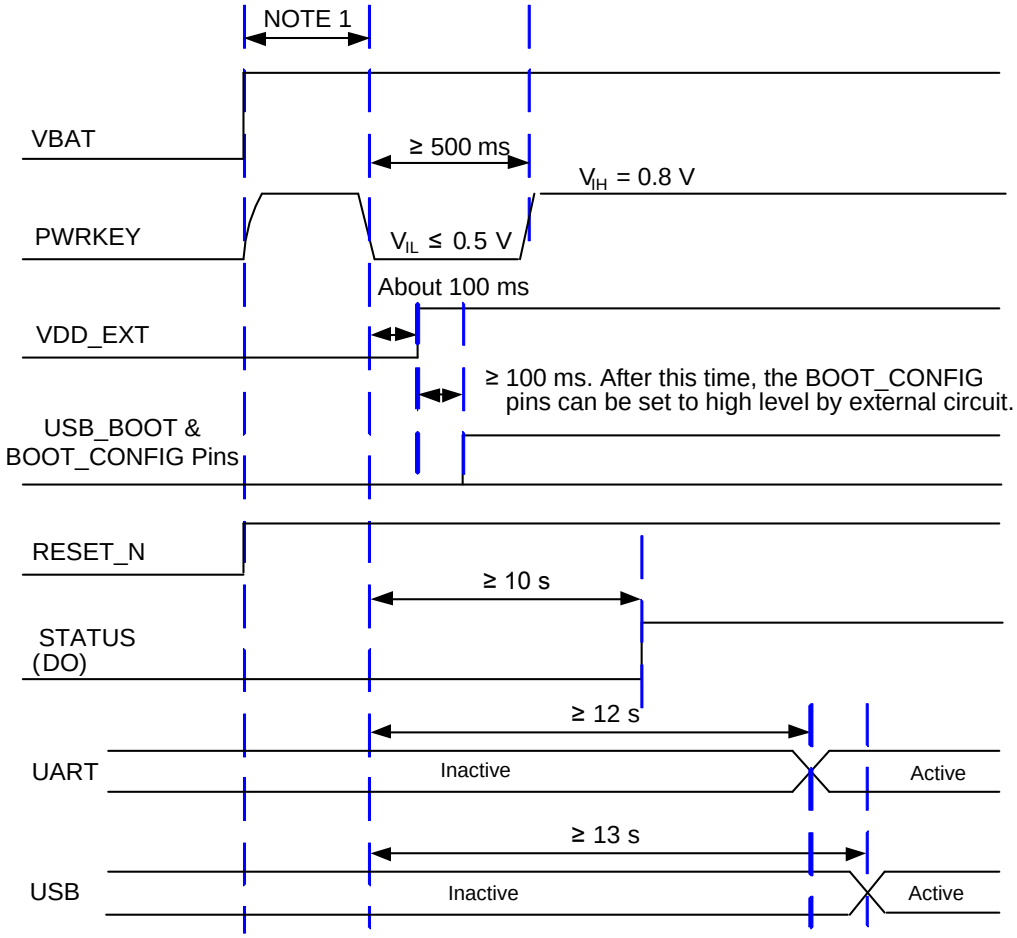


Figure 13: Turn-on Timing

NOTE

1. Make sure that VBAT is stable before pulling down PWRKEY pin. It is recommended that the time between powering up VBAT and pulling down PWRKEY pin is not less than 30 ms.
2. PWRKEY can be pulled down directly to GND with a recommended 10 kΩ resistor if the module needs to be powered on automatically and shutdown is not needed.
3. USB_BOOT pin and BOOT_CONFIG pins (SPI_CLK, PCM_CLK, PCM_SYNC) cannot be pulled up before startup.

3.8. Turn-off

Either of the following methods can be used to turn off the module normally:

- Use the PWRKEY pin.
- Use **AT+QPOWD**

3.8.1. Turn-off with PWRKEY

Driving PWRKEY low for at least 650 ms, the module will execute power-down procedure after the PWRKEY is released. The turn-off timing is illustrated in the following figure.

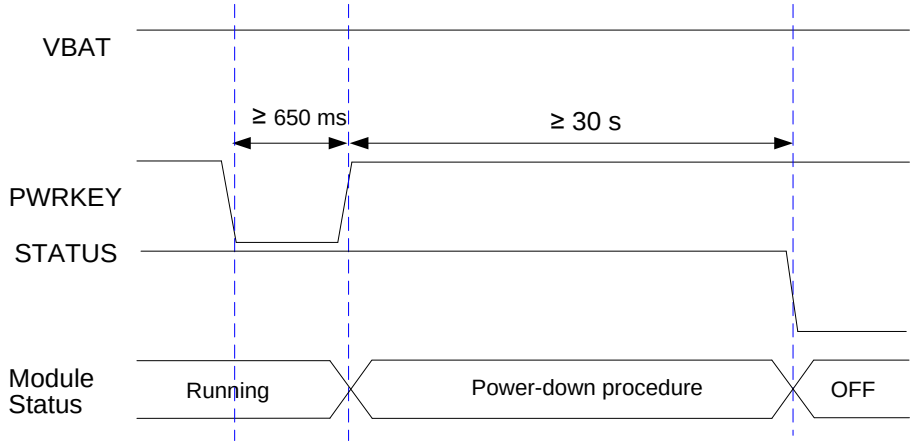


Figure 14: Turn-off Timing

3.8.2. Turn-off with AT Command

It is also a safe way to use **AT+QPOWD** to turn off the module, which is similar to turning off the module via PWRKEY pin. See ***document [3]*** for details about the **AT+QPOWD**.

NOTE

1. To avoid corrupting the data in the internal flash, do not switch off the power supply when the module works normally. Only after the module is turned off by PWRKEY or AT command, the power supply can be cut off.
2. When turning off module with the AT command, keep PWRKEY at high level after the execution of the command. Otherwise, the module will be turned on automatically again after successful turn-off.

3.9. Reset

The module can be reset by driving RESET_N low for 150–460 ms.

Table 8: Pin Definition of RESET_N

Pin Name	Pin No.	Description	Comment
RESET_N	17	Reset the module	Require pull-up resistor to 1.8 V internally. Active low. If unused, keep it open.

The recommended circuit is similar to the PWRKEY control circuit. An open drain/collector driver or button can be used to control the RESET_N.

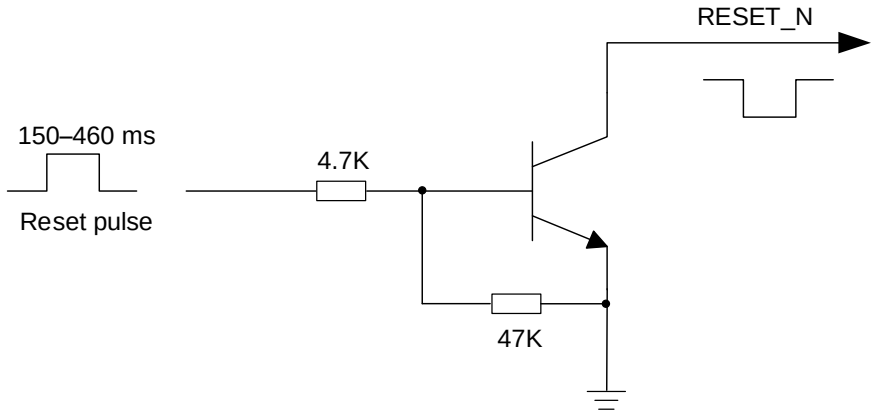


Figure 15: Reference Circuit of RESET_N by Using Driving Circuit

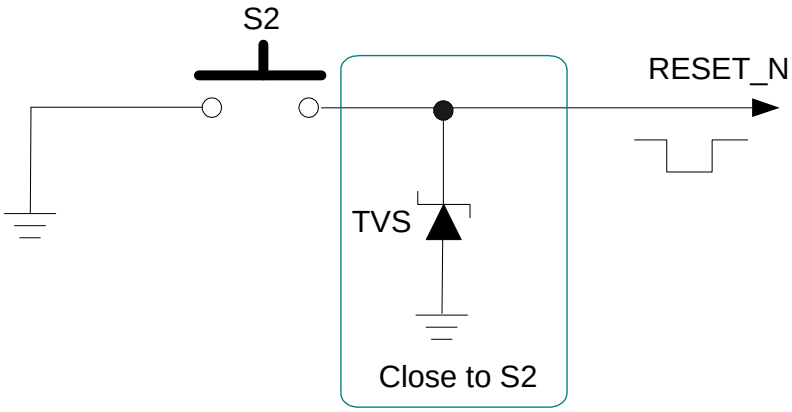


Figure 16: Reference Circuit of RESET_N by Using a Button

The reset timing is illustrated in the following figure.

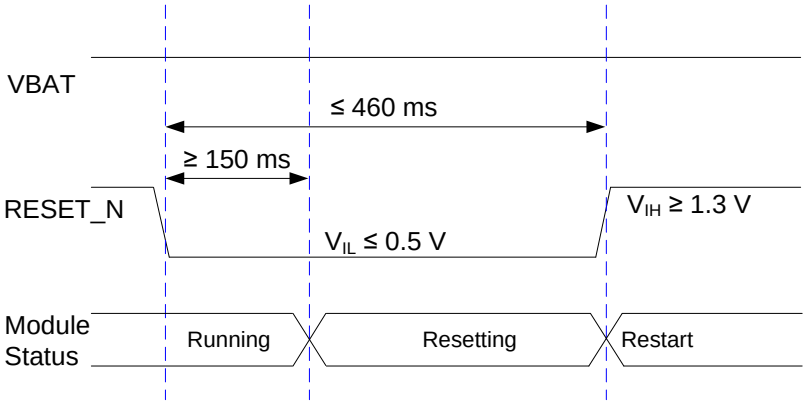


Figure 17: Reset Timing

NOTE

1. Use RESET_N only when turning off the module by **AT+QPOWD** and PWRKEY pin fails.
2. Ensure that there is no large capacitance on PWRKEY and RESET_N pins.

3.10. (U)SIM Interfaces

The module provides two (U)SIM interfaces, and only one (U)SIM card can work at a time. The (U)SIM1 and (U)SIM2 cards can be switched by **AT+QDSIM**. For more details, see ***document [6]***.

The (U)SIM interface circuitry meets ETSI and IMT-2000 requirements. Both 1.8 V and 3.0 V (U)SIM cards are supported.

Table 9: Pin Definition of (U)SIM Interfaces

Pin Name	Pin No.	I/O	Description	Comment
USIM_GND	47	-	Specified ground for (U)SIM card	Connect to ground of (U)SIM card connector.
USIM1_VDD	43	PO	(U)SIM1 card power supply	Either 1.8 V or 3.0 V is supported by the module automatically.
USIM1_DATA	45	DIO	(U)SIM1 card data	
USIM1_CLK	46	DO	(U)SIM1 card clock	
USIM1_RST	44	DO	(U)SIM1 card reset	

USIM1_PRESENCE	42	DI	(U)SIM1 card hot-plug detect	1.8 V power domain. If unused, keep it open.
USIM2_VDD	87	PO	(U)SIM2 card power supply	Either 1.8 V or 3.0 V is supported by the module automatically. If unused, keep it open.
USIM2_DATA	86	DIO	(U)SIM2 card data	If unused, keep them open.
USIM2_CLK	84	DO	(U)SIM2 card clock	
USIM2_RST	85	DO	(U)SIM2 card reset	
USIM2_PRESENCE	83	DI	(U)SIM2 card hot-plug detect	1.8 V power domain. If unused, keep it open.

The module supports (U)SIM card hot-plug via USIM_PRESENCE pin, and both high-level and low-level detection are supported. The function is disabled by default and can be enabled via **AT+QSIMDET**. See ***document [3]*** for more details about **AT+QSIMDET**.

The following figure shows a reference design for (U)SIM interface with an 8-pin (U)SIM card connector.

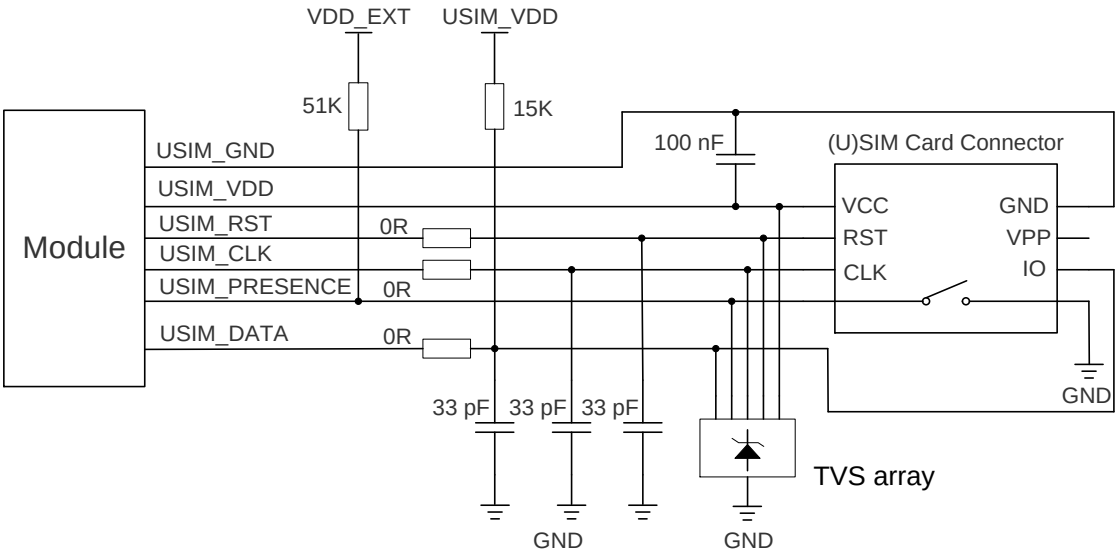


Figure 18: Reference Circuit of (U)SIM Interface with an 8-pin (U)SIM Card Connector

If (U)SIM card detection function is not needed, keep USIM_PRESENCE unconnected. A reference circuit of (U)SIM interface with a 6-pin (U)SIM card connector is illustrated in the following figure.

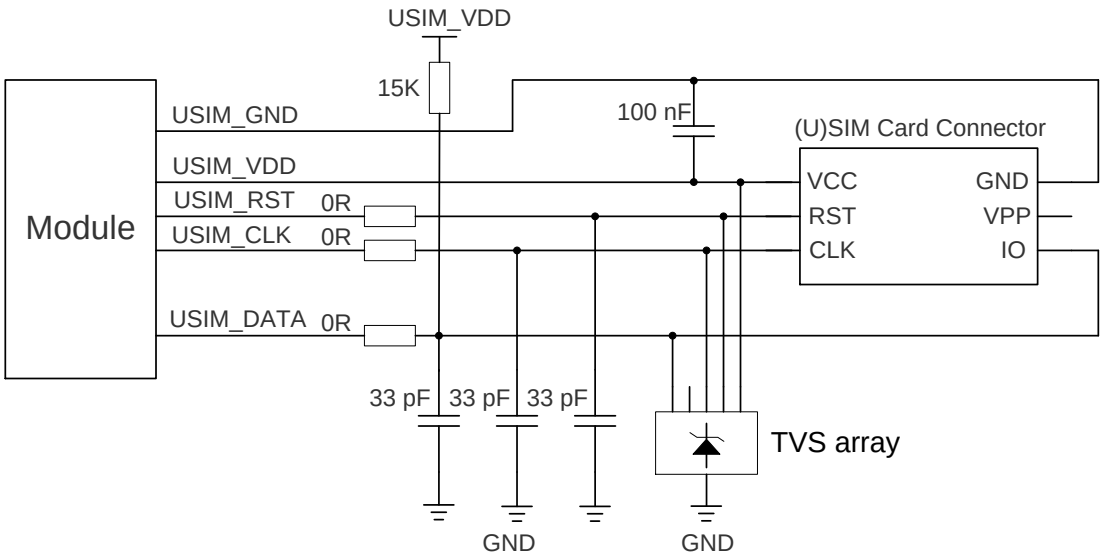


Figure 19: Reference Circuit of (U)SIM Interface with a 6-pin (U)SIM Card Connector

To enhance the reliability and availability of the (U)SIM cards in your applications, follow the criteria below in the (U)SIM circuit design:

- Keep placement of (U)SIM card connector to the module as close as possible. Keep the trace length as short as possible, at most 200 mm.
- Keep (U)SIM card signals away from RF and power supply traces.
- Make sure the bypass capacitor between USIM_VDD and USIM_GND not exceeding 1 μF, and place it as close to (U)SIM card connector as possible. If the ground is complete on your PCB, USIM_GND can be connected to PCB ground directly.
- To avoid cross-talk between USIM_DATA and USIM_CLK, keep them away from each other and shield them with surrounded ground.
- For better ESD protection, it is recommended to add a TVS array whose parasitic capacitance should be less than 15 pF. The 0 Ω resistors should be added in series between the module and the (U)SIM card to facilitate debugging. Additionally, add 33 pF capacitors in parallel among USIM_DATA, USIM_CLK and USIM_RST signal traces to filter out RF interference. Note that the (U)SIM peripheral circuit should be close to the (U)SIM card connector.
- The pull-up resistor on USIM_DATA trace can improve anti-jamming capability when long layout trace and sensitive occasion are applied, and should be placed close to the (U)SIM card connector.

3.11. USB Interface

The module contains one integrated Universal Serial Bus (USB) interface which complies with the USB 2.0 specification and supports high-speed (480 Mbps) and full-speed (12 Mbps) modes.

The USB interface can only serve as a slave device and is used for AT command communication, data transmission, GNSS NMEA sentence output, software debugging, firmware upgrade and voice over USB.

The following table shows the pin definition of USB interface.

Table 10: Pin Definition of USB Interface

Pin Name	Pin No.	I/O	Description	Comment
USB_DP	9	AIO	USB differential data (+)	USB 2.0 compliant. Require differential impedance of 90 Ω.
USB_DM	10	AIO	USB differential data (-)	
USB_VBUS	8	AI	USB connection detect	Typical value: 5.0 V If unused, keep it open.
GND	3	-	Ground	

For more details about USB 2.0 specifications, visit http://www.usb.org/home.

The test points are recommended to be reserved for firmware upgrade and software debugging in your design. The following figure shows a reference circuit of USB interface.

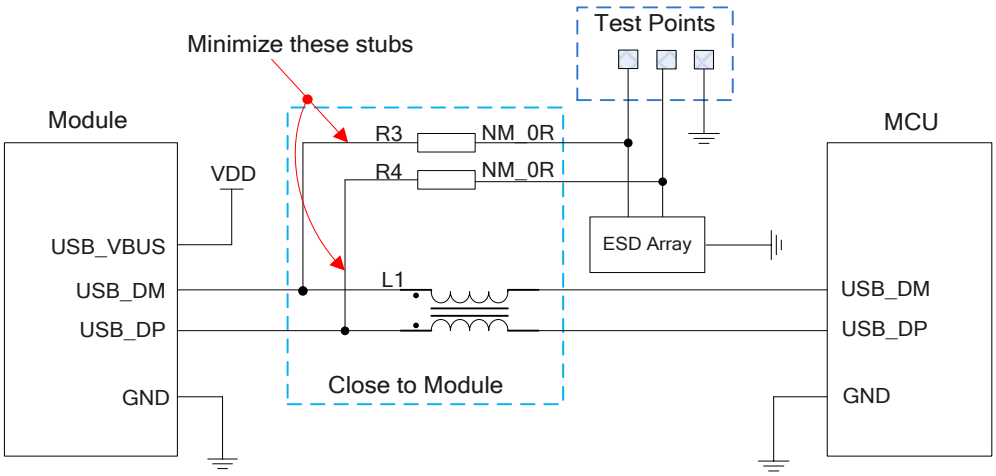


Figure 20: Reference Circuit of USB Interface

A common mode choke L1 is recommended to be added in series between the module and MCU to suppress EMI. Meanwhile, the 0 Ω resistors (R3 and R4) should be added in series between the module and the test points to facilitate debugging, and the resistors are not mounted by default. To ensure the integrity of USB data trace signal, L1, R3 and R4 components must be placed close to the module, and also these resistors should be placed close to each other. The extra stubs of trace must be as short as possible.

To meet USB 2.0 specification, the following principles should be complied with when designing the USB interface.

- It is important to route the USB signal traces as differential pairs with ground surrounded. The impedance of USB differential trace is 90 Ω.
- Do not route signal traces under crystals, oscillators, magnetic devices and RF signal traces. It is important to route the USB differential traces in inner-layer of the PCB, and surround the traces with ground on that layer and with ground planes above and below.
- Junction capacitance of the ESD protection components might cause influences on USB data traces, so pay attention to the selection of the components. Typically, the stray capacitance should not exceed 2 pF.
- Keep the ESD protection components as close to the USB connector as possible.

3.12. UART Interfaces

The module provides two UART interfaces: the main UART and the debug UART. The following shows their features.

- The main UART interface supports 9600 bps, 19200 bps, 38400 bps, 57600 bps, 115200 bps, 230400 bps, 460800 bps and 921600 bps baud rates, and the default is 115200 bps. It supports RTS and CTS hardware flow control, and is used for AT command communication and data transmission.
- The debug UART interface supports 115200 bps baud rate. It is used for Linux console and log output.

The following tables show the pin definition of the two UART interfaces.

Table 11: Pin Definition of Main UART Interface

Pin Name	Pin No.	I/O	Description	Comment
RI	39	DO	Ring indication	1.8 V power domain. If unused, keep them open.
DCD	38	DO	Data carrier detect	
CTS	36	DO	Clear to send signal from the module	Connect to MCU's CTS. 1.8 V power domain. If unused, keep it open.
RTS	37	DI	Request to send signal to the module	Connect to MCU's RTS. 1.8 V power domain. If unused, keep it open.

DTR	30	DI	Data terminal ready; Sleep mode control	1.8 V power domain. Pulled up by default. Low level wakes up the module. If unused, keep it open.
TXD	35	DO	Transmit	1.8 V power domain. If unused, keep them open.
RXD	34	DI	Receive	

Table 12: Pin Definition of Debug UART Interface

Pin Name	Pin No.	I/O	Description	Comment
DBG_TXD	23	DO	Debug UART transmit	1.8 V power domain. If unused, keep them open.
DBG_RXD	22	DI	Debug UART receive	

The module provides 1.8 V UART interfaces. A voltage-level translator should be used if your application is equipped with a 3.3 V UART interface. A voltage-level translator TXS0108EPWR provided by Texas Instruments is recommended. The following figure shows a reference design.

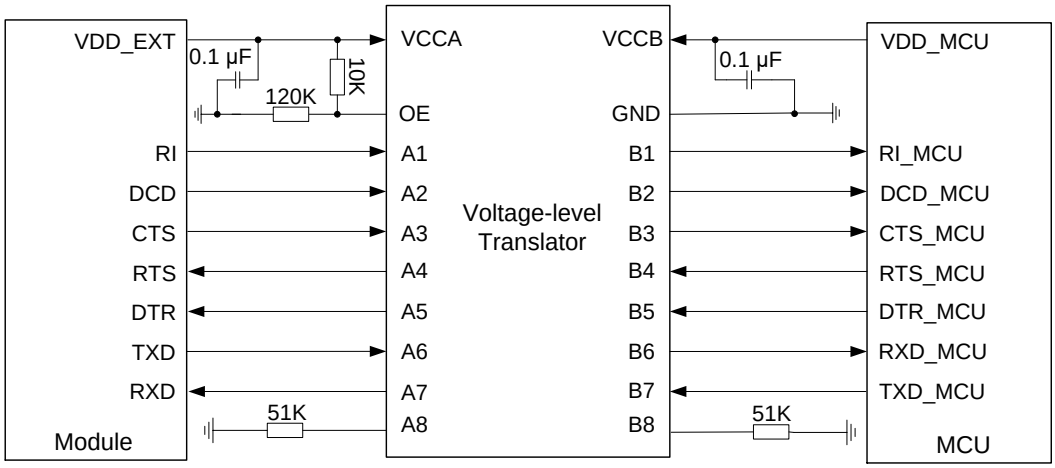


Figure 21: Reference Circuit with Voltage-level Translator

Visit http://www.ti.com for more information.

Another example with transistor circuit is shown as below. For the design of circuits shown in dotted lines, see that shown in solid lines, but pay attention to the direction of connection.

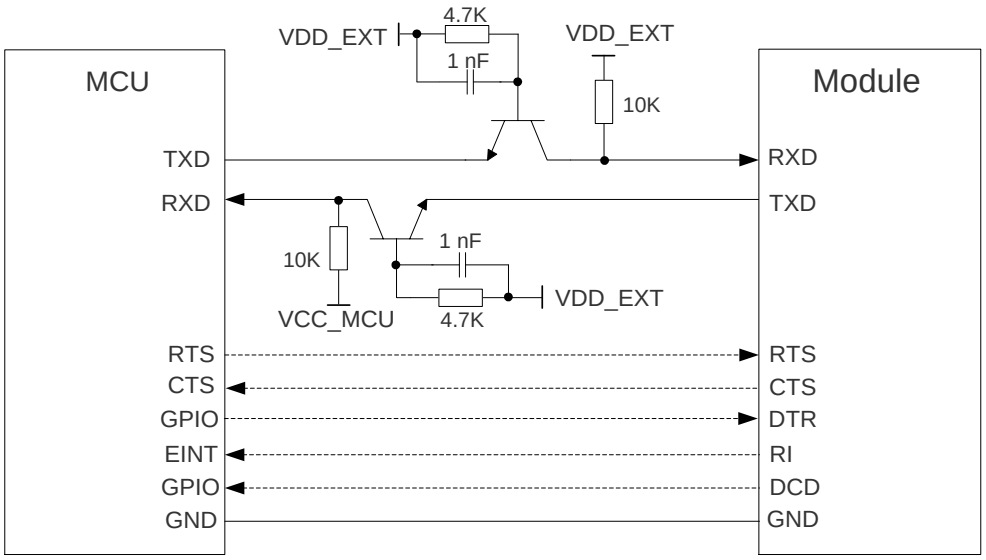


Figure 22: Reference Circuit with Transistor Circuit

NOTE

1. Transistor circuit solution is not suitable for applications with high baud rates exceeding 460 kbps.
2. Please note that the module's CTS is connected to the MCU's CTS, and the module's RTS is connected to the MCU's RTS.
3. To increase the stability of UART communication, it is recommended to add UART hardware flow control design.

3.13. PCM and I2C Interfaces

The module provides one Pulse Code Modulation (PCM) digital interface for audio design which supports the following modes, and provides one I2C interface:

- Primary mode (short frame synchronization, works as both master and slave)
- Auxiliary mode (long frame synchronization, works as master only)

In primary mode, the data is sampled on the falling edge of the PCM_CLK and transmitted on its rising edge. The PCM_SYNC falling edge represents the MSB. In this mode, the PCM interface supports 256 kHz, 512 kHz, 1024 kHz or 2048 kHz PCM_CLK at 8 kHz PCM_SYNC, and also supports 4096 kHz PCM_CLK at 16 kHz PCM_SYNC.

In auxiliary mode, the data is also sampled on the falling edge of the PCM_CLK and transmitted on its rising edge. The PCM_SYNC rising edge represents the MSB. In this mode, the PCM interface operates with a 256 kHz, 512 kHz, 1024 kHz or 2048 kHz PCM_CLK and an 8 kHz, 50 % duty cycle PCM_SYNC.

The module supports 16-bit linear data format. The following figures show the primary mode's timing relationship with 8 kHz PCM_SYNC and 2048 kHz PCM_CLK, as well as the auxiliary mode's timing relationship with 8 kHz PCM_SYNC and 256 kHz PCM_CLK.

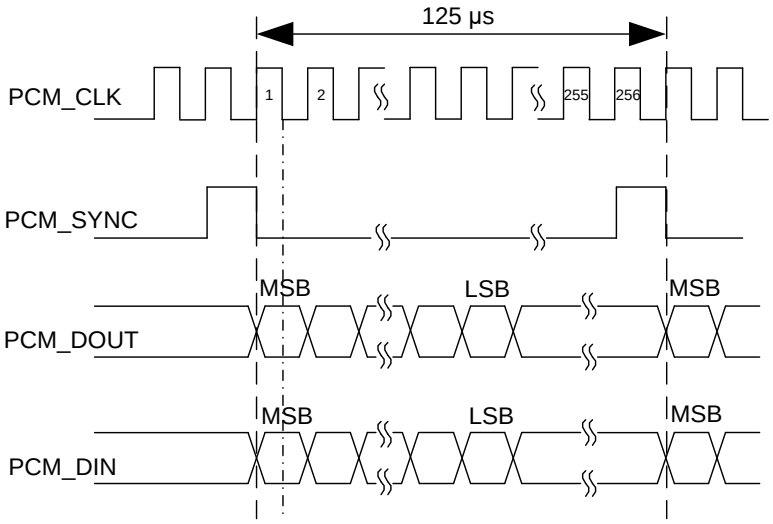


Figure 23: Primary Mode Timing

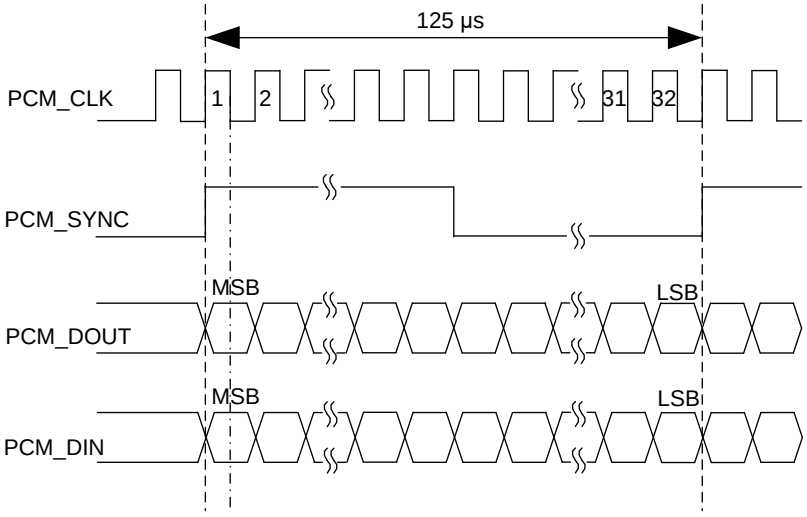


Figure 24: Auxiliary Mode Timing

The following table shows the pin definition of PCM and I2C interfaces which can be applied on audio codec design.

Table 13: Pin Definition of PCM and I2C Interfaces

Pin Name	Pin No.	I/O	Description	Comment
PCM_DIN	6	DI	PCM data input	1.8 V power domain. If unused, keep them open.
PCM_DOUT	7	DO	PCM data output	
PCM_SYNC	5	DIO	PCM data frame sync	1.8 V power domain. In master mode, they are output signals. In slave mode, they are input signals. If unused, keep them open.
PCM_CLK	4	DIO	PCM data clock	
I2C_SCL	40	OD	I2C serial clock (for external codec)	Require an external pull-up to 1.8 V If unused, keep them open.
I2C_SDA	41	OD	I2C serial data (for external codec)	

The clock and mode can be configured by AT command, and the default configuration is master mode using short frame synchronization format with 2048 kHz PCM_CLK and 8 kHz PCM_SYNC. See ***document [3]*** about **AT+QDAI** for details.

The following figure shows a reference design of PCM and I2C interfaces with external codec IC.

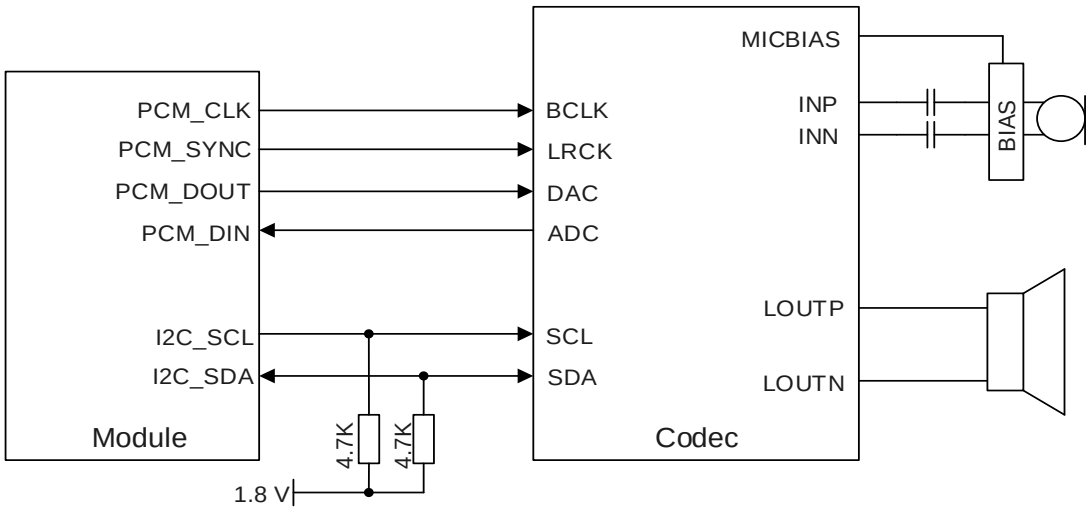


Figure 25: Reference Circuit of PCM Application with Audio Codec

NOTE

1. It is recommended to reserve an RC (R = 22 Ω, C = 22 pF) circuit on the PCM traces, especially for PCM_CLK.
2. The module works as a master device pertaining to I2C interface.

3.14. SPI

SPI only acts as master device. It provides a duplex, synchronous and serial communication link with the peripheral devices. It is dedicated to one-to-one connection without chip selection. Its operation voltage is 1.8 V with clock rates up to 50 MHz.

The following table shows the pin definition of SPI.

Table 14: Pin Definition of SPI

Pin Name	Pin No.	I/O	Description	Comment
SPI_CLK	26	DO	SPI clock	1.8 V power domain. If unused, keep them open.
SPI_MOSI	27	DO	SPI master-out slave-in	
SPI_MISO	28	DI	SPI master-in slave-out	

The following figure shows a reference design of SPI with a peripheral.

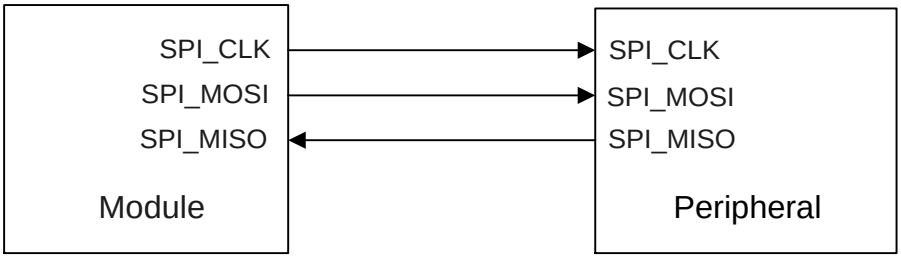


Figure 26: Reference Circuit of SPI with Peripheral

NOTE

The module provides 1.8 V SPI. A voltage-level translator should be used between the module and the host if your application is equipped with a 3.3 V processor or device interface.

3.15. Indication Signals

3.15.1. Network Status Indication

The module provides one network indication pin: NETLIGHT. The pin is used to indicate LED status. The following tables describe the pin definition and logic level changes of NETLIGHT in different network status.

Table 15: Pin Definition of Network Status Indication

Pin Name	Pin No.	I/O	Description	Comment
NETLIGHT	21	DO	Indicate the module's network activity status	1.8 V power domain. If unused, keep it open.

Table 16: Working State of Network Status Indication

Pin Name	Logic Level Change	Network Status
NETLIGHT	Blink slowly (200 ms High/1800 ms Low)	Network searching
	Blink slowly (1800 ms High/200 ms Low)	Idle
	Blink quickly (125 ms High/125 ms Low)	Data transmission is ongoing
	Always High	Voice calling

A reference circuit is shown in the following figure.

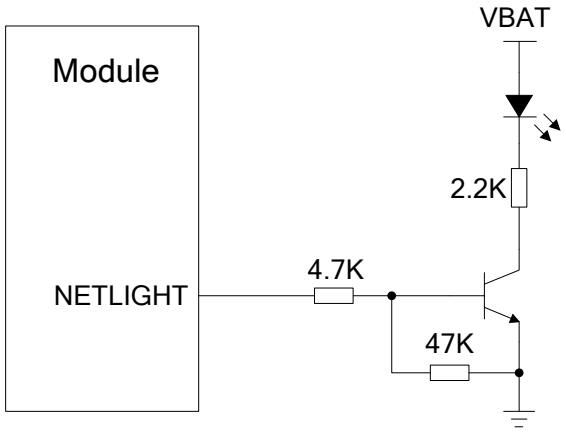


Figure 27: Reference Circuit of Network Status Indication

3.15.2. STATUS

The STATUS pin is set as the module's operating status indicator. It will output high level when the module is turned on normally. The following table describes the pin definition of STATUS.

Table 17: Pin Definition of STATUS

Pin Name	Pin No.	I/O	Description	Comment
STATUS	20	DO	Indicate the module's operating status	1.8 V power domain. If unused, keep it open.

The following figure shows the reference circuit of STATUS.

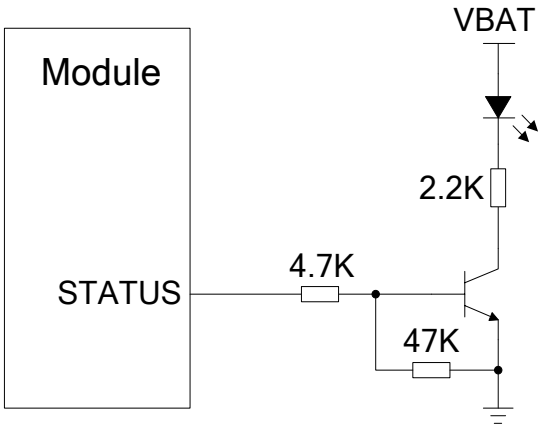


Figure 28: Reference Circuit of STATUS

3.15.3. RI

AT+QCFG="risignaltype","physical" can be used to configure RI behaviors. See ***document [4]*** for details.

No matter on which port (UART port, USB AT port or USB modem port) a URC is presented, URC will trigger the behaviors of RI pin.

NOTE

URC can be outputted from UART port, USB AT port or USB modem port through configuration via **AT+QURCCFG**. The default port is USB AT port. See ***document [3]*** for details.

The default behaviors of the RI are shown as below, and can be changed by **AT+QCFG="urc/ri/ring"**. See ***document [4]*** for details.

Table 18: Default Behaviors of RI

State	Response
Idle	RI keeps at high level.
URC	RI outputs 120 ms low pulse when a new URC returns.

3.16. ADC Interface

The module provides one analog-to-digital converter (ADC) interface. **AT+QADC=0** can be used to read the voltage value on ADC0 pin. For more details about the command, see ***document [3]***.

To improve the accuracy of ADC voltage values, the traces of ADC should be surrounded by ground.

Table 19: Pin Definition of ADC Interface

Pin Name	Pin No.	I/O	Description	Comment
ADC0	24	AI	General-purpose ADC interface	If unused, keep it open.

The following table describes the characteristics of ADC interface.

Table 20: Characteristics of ADC Interface

Parameter	Min.	Typ.	Max.	Unit
ADC0 Input Voltage Range	0.3	-	VBAT_BB	V
ADC Resolution	-	-	15	bits

NOTE

1. It is prohibited to supply any voltage to ADC pin when ADC pin is not powered by VBAT.
2. It is recommended to use resistor divider circuit for ADC application.

3.17. USB_BOOT Interface

The module provides a USB_BOOT pin. You can pull up USB_BOOT to VDD_EXT before VDD_EXT is powered up, and the module will enter forced download mode when it is turned on. In this mode, the module supports firmware upgrade over USB interface.

Table 21: Pin Definition of USB_BOOT Interface

Pin Name	Pin No.	I/O	Description	Comment
USB_BOOT	75	DI	Force the module to enter download mode	1.8 V power domain. Active high. A test point is recommended to be reserved.

The following figures show the reference circuit of the USB_BOOT interface and the timing of entering forced download mode.

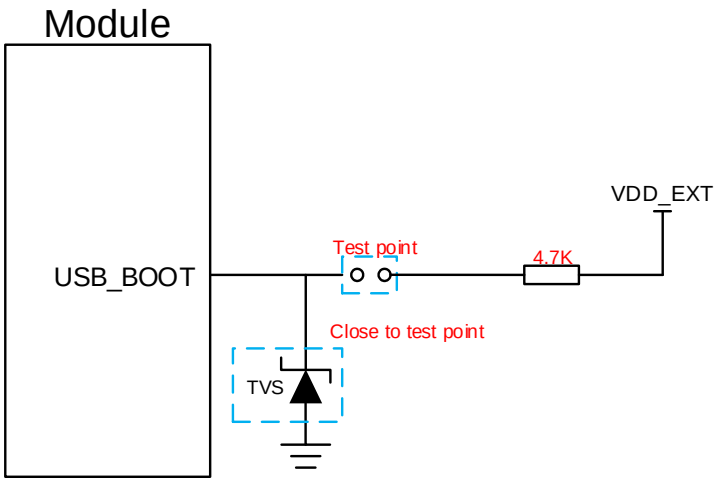


Figure 29: Reference Circuit of USB_BOOT Interface

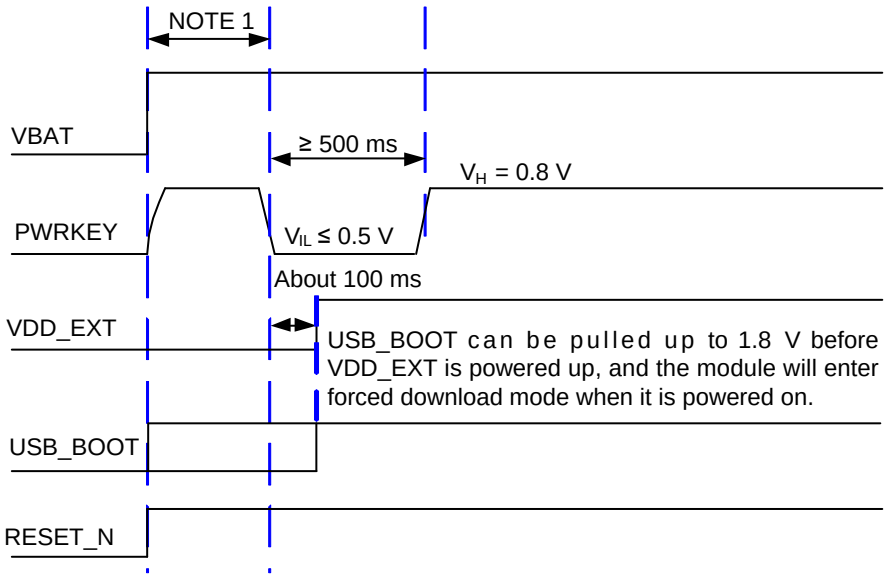


Figure 30: Timing of Entering Forced Download Mode

NOTE

1. Make sure that VBAT is stable before pulling down PWRKEY pin. It is recommended that the time between powering up VBAT and pulling down PWRKEY pin is not less than 30 ms.
2. When using MCU to control the module to enter the forced download mode, follow the above timing sequence. It is not recommended to pull up USB_BOOT to 1.8 V before powering up VBAT. Short the test points as shown in ***Figure 29*** to manually force the module to enter download mode.

4 RF Specifications

The impedance of the antenna port is 50 Ω. The antenna interfaces of the modules are shown as follow:

- One main antenna interface
- One Rx-diversity antenna interface (which is used to resist the fall of signals caused by high-speed movement and multipath effect)
- One GNSS antenna interface (only EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX supports the interface)

Appropriate antenna type and design should be used with matched antenna parameters according to specific application. It is required to perform a comprehensive functional test for the RF design before mass production of terminal products. The entire content of this chapter is provided for illustration only. Analysis, evaluation and determination are still necessary when designing target products.

4.1. Cellular Network

4.1.1. Antenna Interfaces and Frequency Bands

The pin definition of main antenna and Rx-diversity antenna interfaces is shown below.

Table 22: Pin Definition of RF Antennas

Pin Name	Pin No.	I/O	Description	Comment
ANT_MAIN	60	AIO	Main antenna interface	50 Ω characteristic impedance.
ANT_GNSS/ ANT_DIV [13]	49	AI	GNSS antenna interface/ Diversity antenna interface	50 Ω characteristic impedance. If unused, keep them open.
ANT_DIV/ RESERVED [14]	56	AI/-	Diversity antenna interface/ Reserved	

[13] Pin 49 is defined as ANT_GNSS on EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX, while it is defined as ANT_DIV on EG91-E.
[14] Pin 56 is defined as ANT_DIV on EG91-EX/-JP/-NA/-NAX/-NAXD/-VX, while it is defined as RESERVED on EG91-AUX/-E. Rx-diversity antenna is not supported on EG91-AUX.

Table 23: Module Operating Frequencies

3GPP Band	Transmit	Receive	Unit
GSM850	824–849	869–894	MHz
EGSM900	880–915	925–960	MHz
DCS1800	1710–1785	1805–1880	MHz
PCS1900	1850–1910	1930–1990	MHz
WCDMA B1	1920–1980	2110–2170	MHz
WCDMA B2	1850–1910	1930–1990	MHz
WCDMA B4	1710–1755	2110–2155	MHz
WCDMA B5	824–849	869–894	MHz
WCDMA B8	880–915	925–960	MHz
LTE-FDD B1	1920–1980	2110–2170	MHz
LTE FDD B2	1850–1910	1930–1990	MHz
LTE-FDD B3	1710–1785	1805–1880	MHz
LTE FDD B4	1710–1755	2110–2155	MHz
LTE FDD B5	824–849	869–894	MHz
LTE-FDD B7	2500–2570	2620–2690	MHz
LTE-FDD B8	880–915	925–960	MHz
LTE FDD B12	699–716	729–746	MHz
LTE FDD B13	777–787	746–756	MHz
LTE FDD B18	815–830	860–875	MHz
LTE FDD B19	830–845	875–890	MHz
LTE-FDD B20	832–862	791–821	MHz
LTE-FDD B25	1850–1915	1930–1995	MHz
LTE-FDD B26	814–849	859–894	MHz

LTE-FDD B28A	703–733	758–788	MHz
LTE-FDD B28	703–748	758–803	MHz
LTE-FDD B66	1710–1780	2110–2180	MHz

4.1.2. Antenna Tuner Control Interfaces

The module can use GRFC (generic RF control) interfaces to control external antenna tuner.

Table 24: Pin Definition of GRFC Interfaces

Pin Name	Pin No.	I/O	Description	Comment
GRFC9/NC [15]	76	DO/-	Generic RF controller/Not connected	If unused, keep them open.
GRFC10/NC [15]	77	DO/-	Generic RF controller/Not connected	

4.1.3. Tx Power

The following table shows the Tx power of the module.

Table 25: Tx Power

Frequency Band	Max. Tx Power	Min. Tx Power
GSM850/EGSM900	33 dBm ±2 dB	5 dBm ±5 dB
DCS1800/PCS1900	30 dBm ±2 dB	0 dBm ±5 dB
GSM850/EGSM900 (8-PSK)	27 dBm ±3 dB	5 dBm ±5 dB
DCS1800/PCS1900 (8-PSK)	26 dBm ±3 dB	0 dBm ±5 dB
WCDMA bands	23 dBm ±2 dB	< -49 dBm
LTE-FDD bands	23 dBm ±2 dB	< -39 dBm

[15] Pin 76 is defined as GRFC9 on EG91-AUX/-JP, while it is defined as NC on EG91-E/-EX/-NA/-NAX/-NAXD/-VX. Pin 77 is defined as GRFC10 on EG91-AUX/-JP, while it is defined as NC on EG91-E/-EX/-NA/-NAX/-NAXD/-VX.

NOTE

For GPRS transmission on 4 uplink timeslots, the maximum output power reduction is 4.0 dB. The design conforms to 3GPP TS 51.010-1 ***subclause 13.16***.

4.1.4. Receiver Sensitivity

The following tables show the conducted RF receiver sensitivity of the module.

4.1.4.1. EG91-AUX Conducted RF Receiver Sensitivity

Table 26: EG91-AUX Conducted RF Receiver Sensitivity

Frequency Band	Receiver Sensitivity (Typ.)			3GPP (SIMO)
	Primary	Diversity [16]	SIMO	
GSM850	-109.1 dBm	-	-	-102 dBm
EGSM900	-109.7 dBm	-	-	-102 dBm
DCS1800	-110.0 dBm	-	-	-102 dBm
PCS1900	-109.4 dBm	-	-	-102 dBm
WCDMA B1	-109.2 dBm	-	-	-106.7 dBm
WCDMA B2	-109.8 dBm	-	-	-104.7 dBm
WCDMA B5	-110 dBm	-	-	-104.7 dBm
WCDMA B8	-110 dBm	-	-	-103.7 dBm
LTE-FDD B1 (10 MHz)	-97.2 dBm	-	-	-96.3 dBm
LTE-FDD B2 (10 MHz)	-97.7 dBm	-	-	-94.3 dBm
LTE-FDD B3 (10 MHz)	-98.2 dBm	-	-	-93.3 dBm
LTE-FDD B4 (10 MHz)	-97.7 dBm	-	-	-96.3 dBm
LTE-FDD B5 (10 MHz)	-99.2 dBm	-	-	-94.3 dBm
LTE-FDD B7 (10 MHz)	-96.7 dBm	-	-	-94.3 dBm

[16] Rx-diversity antenna is not supported on EG91-AUX.

LTE-FDD B8 (10 MHz)	-98.0 dBm	-	-	-93.3 dBm
LTE-FDD B28 (10 MHz)	-98.7 dBm	-	-	-94.8 dBm
LTE-FDD B66 (10 MHz)	-97.7 dBm	-	-	-95.8 dBm

4.1.4.2. EG91-E Conducted RF Receiver Sensitivity

Table 27: EG91-E Conducted RF Receiver Sensitivity

Frequency Band	Receiver Sensitivity (Typ.)			3GPP (SIMO)
	Primary	Diversity	SIMO	
EGSM900	-108.6 dBm	-	-	-102 dBm
DCS1800	-109.4 dBm	-	-	-102 dBm
WCDMA B1	-109.5 dBm	-110 dBm	-112.5 dBm	-106.7 dBm
WCDMA B8	-109.5 dBm	-110 dBm	-112.5 dBm	-103.7 dBm
LTE-FDD B1 (10 MHz)	-97.5 dBm	-98.3 dBm	-101.4 dBm	-96.3 dBm
LTE-FDD B3 (10 MHz)	-98.3 dBm	-98.5 dBm	-101.5 dBm	-93.3 dBm
LTE-FDD B7 (10 MHz)	-96.3 dBm	-98.4 dBm	-101.3 dBm	-94.3 dBm
LTE-FDD B8 (10 MHz)	-97.1 dBm	-99.1 dBm	-101.2 dBm	-93.3 dBm
LTE-FDD B20 (10 MHz)	-97 dBm	-99 dBm	-101.3 dBm	-93.3 dBm
LTE-FDD B28A (10 MHz)	-98.3 dBm	-99 dBm	-101.4 dBm	-94.8 dBm

4.1.4.3. EG91-EX Conducted RF Receiver Sensitivity

Table 28: EG91-EX Conducted RF Receiver Sensitivity

Frequency Band	Receiver Sensitivity (Typ.)			3GPP (SIMO)
	Primary	Diversity	SIMO	
EGSM900	-109.8 dBm	-	-	-102 dBm

DCS1800	-109.8 dBm	-	-	-102 dBm
WCDMA B1	-110 dBm	-111 dBm	-112.5 dBm	-106.7 dBm
WCDMA B8	-110 dBm	-111 dBm	-112.5 dBm	-103.7 dBm
LTE-FDD B1 (10 MHz)	-98.7 dBm	-98.8 dBm	-102.4 dBm	-96.3 dBm
LTE-FDD B3 (10 MHz)	-98.3 dBm	-99.5 dBm	-102.5 dBm	-93.3 dBm
LTE-FDD B7 (10 MHz)	-97.5 dBm	-98.4 dBm	-100.3 dBm	-94.3 dBm
LTE-FDD B8 (10 MHz)	-98.7 dBm	-99.6 dBm	-102.2 dBm	-93.3 dBm
LTE-FDD B20 (10 MHz)	-97 dBm	-97.5 dBm	-102.2 dBm	-93.3 dBm
LTE-FDD B28 (10 MHz)	-98.2 dBm	-99.5 dBm	-102 dBm	-94.8 dBm

4.1.4.4. EG91-JP Conducted RF Receiver Sensitivity

Table 29: EG91-JP Conducted RF Receiver Sensitivity

Frequency Band	Receiver Sensitivity (Typ.)			3GPP (SIMO)
	Primary	Diversity	SIMO	
LTE-FDD B1 (10 MHz)	-97.7 dBm	-99.3 dBm	-101.4 dBm	-96.3 dBm
LTE-FDD B3 (10 MHz)	-98.4 dBm	-99.2 dBm	-102.1 dBm	-93.3 dBm
LTE-FDD B8 (10 MHz)	-98.3 dBm	-99.8 dBm	-102.1 dBm	-93.3 dBm
LTE-FDD B18 (10 MHz)	-99.2 dBm	-100 dBm	-102.7 dBm	-96.3 dBm
LTE-FDD B19 (10 MHz)	-99.3 dBm	-100.3 dBm	-102.8 dBm	-96.3 dBm
LTE-FDD B26 (10 MHz)	-98.9 dBm	-99.9 dBm	-102.6 dBm	-93.8 dBm

4.1.4.5. EG91-NA Conducted RF Receiver Sensitivity

Table 30: EG91-NA Conducted RF Receiver Sensitivity

Frequency Band	Receiver Sensitivity (Typ.)			3GPP (SIMO)
	Primary	Diversity	SIMO	
WCDMA B2	-110 dBm	-110 dBm	-112.5 dBm	-104.7 dBm
WCDMA B4	-110 dBm	-110 dBm	-112.5 dBm	-106.7 dBm
WCDMA B5	-111 dBm	-111 dBm	-113 dBm	-104.7 dBm
LTE-FDD B2 (10 MHz)	-98 dBm	-99 dBm	-102.2 dBm	-94.3 dBm
LTE-FDD B4 (10 MHz)	-97.8 dBm	-99.5 dBm	-102.2 dBm	-96.3 dBm
LTE-FDD B5 (10 MHz)	-99.6 dBm	-100.3 dBm	-103 dBm	-94.3 dBm
LTE-FDD B12 (10 MHz)	-99.5 dBm	-100 dBm	-102.5 dBm	-93.3 dBm
LTE-FDD B13 (10 MHz)	-99.2 dBm	-100 dBm	-102.5 dBm	-93.3 dBm

4.1.4.6. EG91-NAX Conducted RF Receiver Sensitivity

Table 31: EG91-NAX Conducted RF Receiver Sensitivity

Frequency Band	Receiver Sensitivity (Typ.)			3GPP (SIMO)
	Primary	Diversity	SIMO	
WCDMA B2	-110 dBm	-110 dBm	-112.5 dBm	-104.7 dBm
WCDMA B4	-110 dBm	-110 dBm	-112.5 dBm	-106.7 dBm
WCDMA B5	-111 dBm	-111 dBm	-113 dBm	-104.7 dBm
LTE-FDD B2 (10 MHz)	-98 dBm	-99 dBm	-102.2 dBm	-94.3 dBm
LTE-FDD B4 (10 MHz)	-97.8 dBm	-99.5 dBm	-102.2 dBm	-96.3 dBm
LTE-FDD B5 (10 MHz)	-99.4 dBm	-100 dBm	-102.7 dBm	-94.3 dBm
LTE-FDD B12 (10 MHz)	-99.5 dBm	-100 dBm	-102.5 dBm	-93.3 dBm

LTE-FDD B13 (10 MHz)	-99.2 dBm	-100 dBm	-102.5 dBm	-93.3 dBm
LTE-FDD B25 (10 MHz)	-97.6 dBm	-99 dBm	-102.2 dBm	-92.8 dBm
LTE-FDD B26 (10 MHz)	-99.1 dBm	-99.9 dBm	-102.7 dBm	-93.8 dBm

4.1.4.7. EG91-NAXD Conducted RF Receiver Sensitivity

Table 32: EG91-NAXD Conducted RF Receiver Sensitivity

Frequency Band	Receiver Sensitivity (Typ.)			3GPP (SIMO)
	Primary	Diversity	SIMO	
WCDMA B2	-110 dBm	-110 dBm	-112.5 dBm	-104.7 dBm
WCDMA B4	-110 dBm	-110 dBm	-112.5 dBm	-106.7 dBm
WCDMA B5	-111 dBm	-111 dBm	-113 dBm	-104.7 dBm
LTE-FDD B2 (10 MHz)	-98 dBm	-99 dBm	-102.2 dBm	-94.3 dBm
LTE-FDD B4 (10 MHz)	-97.8 dBm	-99.5 dBm	-102.2 dBm	-96.3 dBm
LTE-FDD B5 (10 MHz)	-99.4 dBm	-100 dBm	-102.7 dBm	-94.3 dBm
LTE-FDD B12 (10 MHz)	-99.5 dBm	-100 dBm	-102.5 dBm	-93.3 dBm
LTE-FDD B13 (10 MHz)	-99.2 dBm	-100 dBm	-102.5 dBm	-93.3 dBm
LTE-FDD B25 (10 MHz)	-97.6 dBm	-99 dBm	-102.2 dBm	-92.8 dBm
LTE-FDD B26 (10 MHz)	-99.1 dBm	-99.9 dBm	-102.7 dBm	-93.8 dBm

4.1.4.8. EG91-VX Conducted RF Receiver Sensitivity

Table 33: EG91-VX Conducted RF Receiver Sensitivity

Frequency Band	Receiver Sensitivity (Typ.)			3GPP (SIMO)
	Primary	Diversity	SIMO	
LTE-FDD B4 (10 MHz)	-98.2 dBm	-99.2 dBm	-102.2 dBm	-96.3 dBm

LTE-FDD B13 (10 MHz)	-99.2 dBm	-100 dBm	-102.5 dBm	-93.3 dBm

4.1.5. Reference Design

A reference design of ANT_MAIN and ANT_DIV antenna interfaces is shown as below. A dual L-type circuit should be reserved for better RF performance. The capacitors C1–C4 are not mounted by default. Place the dual L-type components (R1/C1/C2/C5 and R2/C3/C4/C6) as close to the antenna as possible.

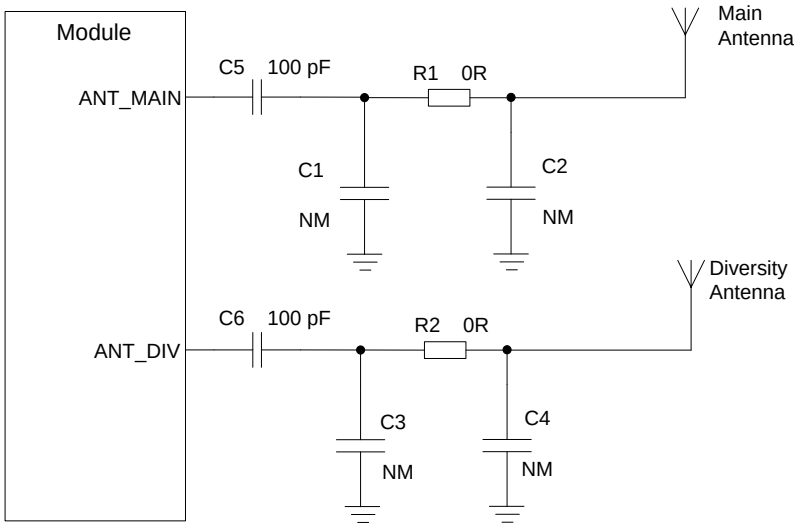


Figure 31: Reference Circuit of RF Antenna Interfaces

NOTE

1. EG91-AUX does not support Rx-diversity.
2. Keep a proper distance between the main antenna and the Rx-diversity antenna to improve the receiver sensitivity.
3. For the operation of ANT_MAIN and ANT_DIV, see **AT+QCFG="divctl"** in ***document [4]*** for more details.
4. Notes on C5 and C6:
   1) If there is DC power at the antenna ports, place capacitors on C5 and C6 to prevent short circuit to ground. The capacitance value is recommended to be 100 pF, which can be adjusted according to the debugging results.
   2) If there is no DC power in the peripheral design:
      a) Do not reserve C5 and C6.
      b) If C5 and C6 have already been reserved, they should be mounted with components, and it is recommended to use 0 Ω resistors. You can also match the components according to the debugging results.

4.2. GNSS (Optional)

The module includes a fully integrated global navigation satellite system solution that supports GPS, GLONASS, BDS, Galileo and QZSS.

The module supports standard NMEA 0183 protocol, and outputs NMEA sentences at 1 Hz data update rate via USB interface by default.

By default, the module's GNSS engine is switched off. It has to be switched on via AT command. For more details about GNSS engine technology and configurations, see ***document [1]***.

4.2.1. Antenna Interfaces and Frequency Bands

The GNSS antenna interface is only supported on EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX. The following tables show pin definition and frequency specification of GNSS antenna interface.

Table 34: Pin Definition of GNSS Antenna Interface

Pin Name	Pin No.	I/O	Description	Comment
ANT_GNSS/ ANT_DIV [17]	49	AI	GNSS antenna interface/ Diversity antenna interface	50 Ω characteristic impedance. If unused, keep it open.

Table 35: GNSS Frequency

GNSS Constellation Type	Frequency	Unit
GPS	1575.42 ±1.023	MHz
GLONASS	1601.7 ±4.2	MHz
Galileo	1575.42 ±2.046	MHz
BDS	1561.098 ±2.046	MHz
QZSS	1575.42 ±1.023	MHz

[17] Pin 49 is defined as ANT_GNSS on EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX, while it is defined as ANT_DIV on EG91-E.

4.2.2. GNSS Performance

The following table shows GNSS performance of the module.

Table 36: GNSS Performance

Parameter	Description	Condition	Typ.	Unit
Sensitivity	Acquisition	Autonomous	-146	dBm
	Reacquisition	Autonomous	-157	dBm
	Tracking	Autonomous	-157	dBm
TTFF	Cold start @ open sky	Autonomous	34.6	s
		AGNSS start	11.57	s
	Warm start @ open sky	Autonomous	26.09	s
		AGNSS start	3.7	s
	Hot start @ open sky	Autonomous	1.8	s
		AGNSS start	3.4	s
Accuracy	CEP-50	Autonomous @ open sky	2.5	m

NOTE

1. Tracking sensitivity: the minimum GNSS signal power at which the module can maintain lock (keep positioning for at least 3 minutes continuously).
2. Reacquisition sensitivity: the minimum GNSS signal power required for the module to maintain lock within 3 minutes after loss of lock.
3. Acquisition sensitivity: the minimum GNSS signal power at which the module can fix position successfully within 3 minutes after executing cold start command.

4.2.3. Reference Design

In any case, it is recommended to use a passive antenna. However, if an active antenna is needed in your application, it is recommended to reserve a π-type attenuation circuit and use a high-performance LDO in the power system design.

A reference design of GNSS antenna is shown as below.

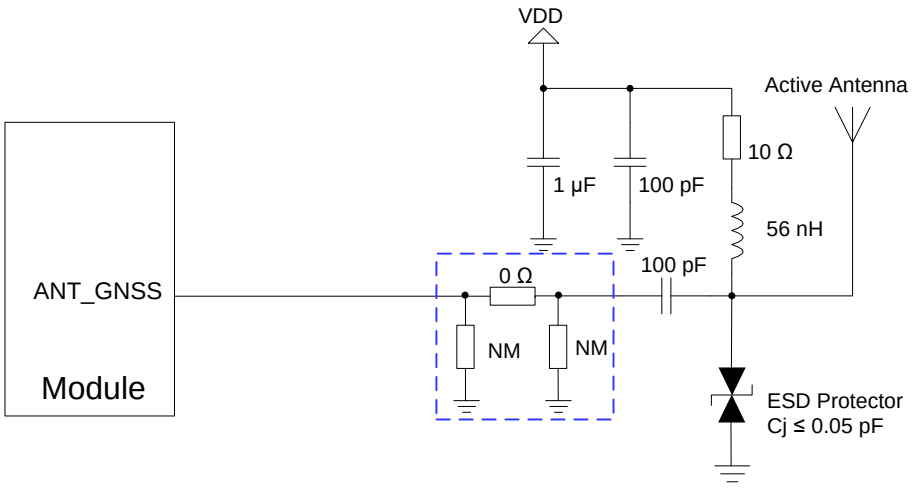


Figure 32: Reference Circuit of GNSS Antenna

NOTE

1. An external LDO can be selected to supply power according to the active antenna requirement.
2. If the module is designed with a passive antenna, you will not need the VDD circuit.
3. It is recommended to use passive antenna. If active antenna or an additional LNA is needed, please contact Quectel Technical Support.

4.2.4. Layout Guidelines

The following layout guidelines should be taken into account in your design.

- Maximize the distance among GNSS antenna, main antenna and Rx-diversity antenna [18].
- Digital circuits such as (U)SIM card, USB interface, camera module and display connector should be kept away from the antennas.
- Use ground vias around the GNSS trace and sensitive analog signal traces to provide coplanar isolation and protection.
- Keep the characteristic impedance for ANT_GNSS trace as 50 Ω.

[18] EG91-AUX does not support Rx-diversity.

4.3. RF Routing Guidelines

For user's PCB, the characteristic impedance of all RF traces should be controlled to 50 Ω. The impedance of the RF traces is usually determined by the trace width (W), the materials' dielectric constant, the height from the reference ground to the signal layer (H), and the spacing between RF traces and grounds (S). Microstrip or coplanar waveguide is typically used in RF layout to control characteristic impedance. The following are reference designs of microstrip or coplanar waveguide with different PCB structures.

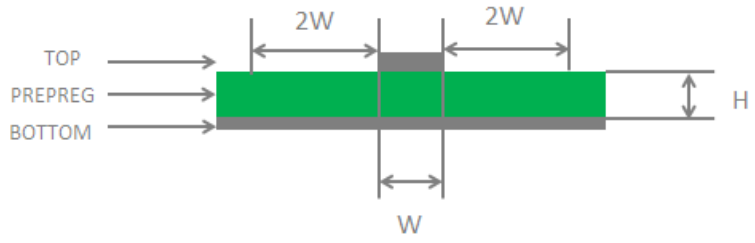


Figure 33: Microstrip Design on a 2-layer PCB

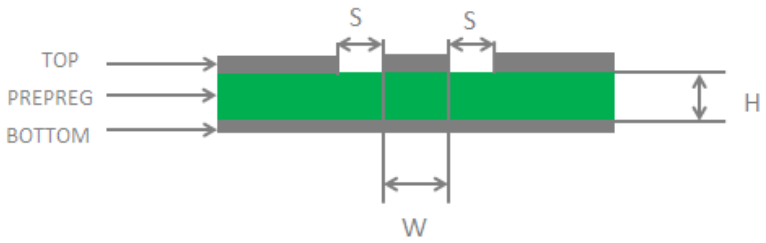


Figure 34: Coplanar Waveguide Design on a 2-layer PCB

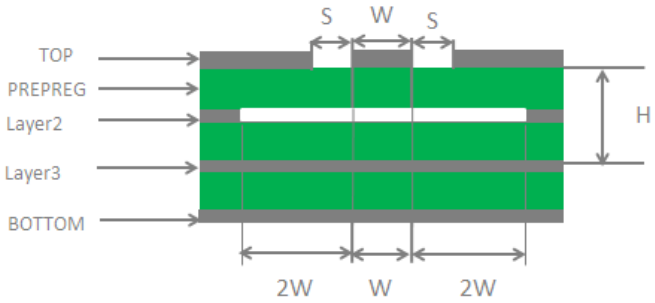


Figure 35: Coplanar Waveguide Design on a 4-layer PCB (Layer 3 as Reference Ground)

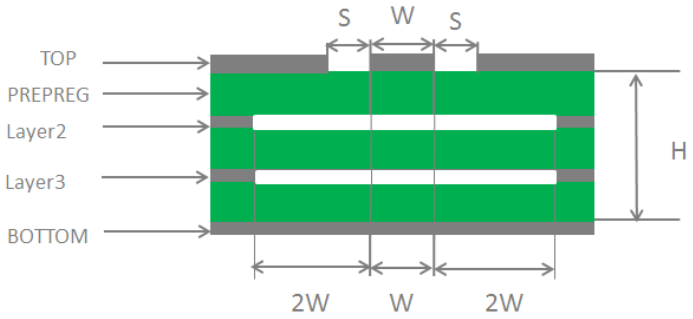


Figure 36: Coplanar Waveguide Design on a 4-layer PCB (Layer 4 as Reference Ground)

To ensure RF performance and reliability, follow the principles below in RF layout design:

- Use an impedance simulation tool to accurately control the characteristic impedance of RF traces to 50 Ω.
- The GND pins adjacent to RF pins should not be designed as thermal relief pads, and should be fully connected to ground.
- The distance between the RF pins and the RF connector should be as short as possible and all the right-angle traces should be changed to curved ones. The recommended trace angle is 135°.
- There should be clearance under the signal pin of the antenna connector or solder joint.
- The reference ground of RF traces should be complete. Meanwhile, adding some ground vias around RF traces and the reference ground could help to improve RF performance. The distance between the ground vias and RF traces should be at least twice the width of RF signal traces (2 × W).
- Keep RF traces away from interference sources (such as DC-DC, (U)SIM/USB/SDIO high frequency digital signals, display signals, and clock signals), and avoid intersection and paralleling between traces on adjacent layers.

For more details about RF layout, refer to ***document [7]***.

4.4. Antenna Design Requirements

The following table shows the requirements on main antenna, Rx-diversity antenna [19] and GNSS antenna.

[19] EG91-AUX does not support Rx-diversity.

Table 37: Antenna Requirements

Type	Requirements
GNSS (Optional)	● Frequency range: 1559–1609 MHz ● Polarization: RHCP or linear ● VSWR: ≤ 2 (Typ.)
	For passive antenna usage: ● Passive antenna gain: > 0 dBi
	For active antenna usage: ● Active antenna noise figure: < 1.5 dB ● Active antenna embedded LNA gain: < 17 dB
Cellular	● VSWR: ≤ 2 ● Efficiency: > 30 % ● Gain: 1 dBi ● Max. input power: 50 W ● Input impedance: 50 Ω ● Vertical polarization ● Cable insertion loss: – **< 1 dB:** LB (<1 GHz) – **< 1.5 dB:** MB (1–2.3 GHz) – **< 2 dB:** HB (> 2.3 GHz)

4.5. RF Connector Recommendation

If RF connector is used for antenna connection, it is recommended to use U.FL-R-SMT connector provided by Hirose.

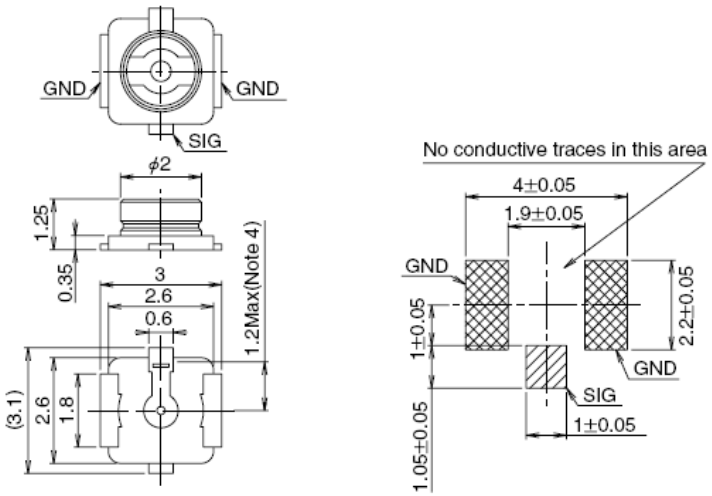


Figure 37: Dimensions of the Receptacle (Unit: mm)

U.FL-LP series mated plugs listed in the following figure can be used to match the U.FL-R-SMT connector.

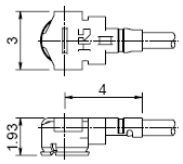
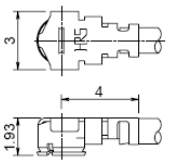
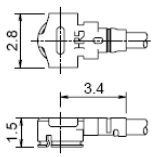
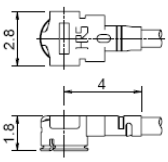
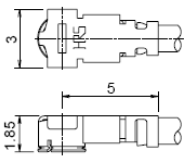
Part No.	U.FL-LP-040	U.FL-LP-066	U.FL-LP(V)-040	U.FL-LP-062	U.FL-LP-088
					
Mated Height	2.5mm Max. (2.4mm Nom.)	2.5mm Max. (2.4mm Nom.)	2.0mm Max. (1.9mm Nom.)	2.4mm Max. (2.3mm Nom.)	2.4mm Max. (2.3mm Nom.)
Applicable cable	Dia. 0.81mm Coaxial cable	Dia. 1.13mm and Dia. 1.32mm Coaxial cable	Dia. 0.81mm Coaxial cable	Dia. 1mm Coaxial cable	Dia. 1.37mm Coaxial cable
Weight (mg)	53.7	59.1	34.8	45.5	71.7
RoHS	YES				

Figure 38: Specifications of Mated Plugs

The following figure describes the space factor of mated connector.

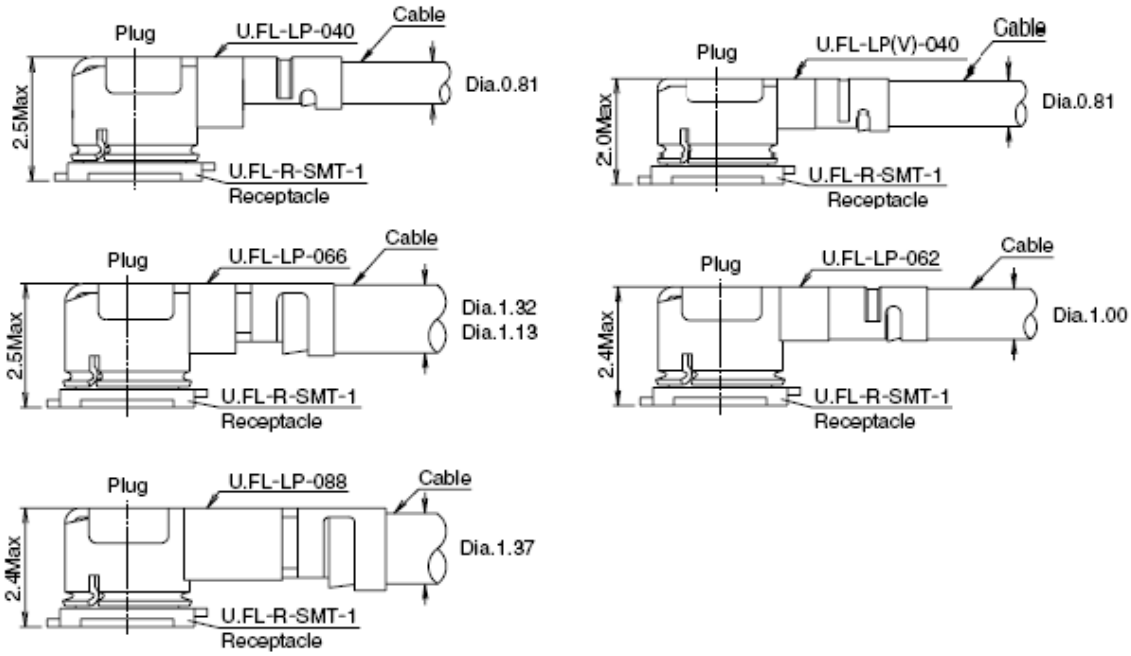


Figure 39: Space Factor of Mated Connectors (Unit: mm)

For more details, visit http://www.hirose.com.

5 Electrical Characteristics and Reliability

5.1. Absolute Maximum Ratings

Absolute maximum ratings for power supply and voltage on digital and analog pins of the module are listed in the following table.

Table 38: Absolute Maximum Ratings

Parameter	Min.	Max.	Unit
VBAT_RF/VBAT_BB	-0.3	4.7	V
USB_VBUS	-0.3	5.5	V
Peak Current of VBAT_BB	-	0.8	A
Peak Current of VBAT_RF	-	1.8	A
Voltage at Digital Pins	-0.3	2.3	V
Input Voltage at ADC0	0	VBAT_BB	V

5.2. Power Supply Ratings

Table 39: Power Supply Ratings

Parameter	Description	Condition	Min.	Typ.	Max.	Unit
VBAT	VBAT_BB and VBAT_RF	The actual input voltages must be kept between the minimum and maximum	3.3	3.8	4.3	V

		values.				
	Voltage drop during burst transmission	At maximum power control level	-	-	400	mV
I_{VBAT}	Peak supply current	At maximum power control level	-	-	2.0	A
USB_VBUS	USB connection detection	-	3.0	5.0	5.25	V

5.3. Operating and Storage Temperatures

The operating and storage temperatures are listed in the following table.

Table 40: Operating and Storage Temperatures

Parameter	Min.	Typ.	Max.	Unit
Normal Operating Temperature [20]	-35	+25	+75	ºC
Extended Operating Temperature [21]	-40	-	+85	ºC
Storage Temperature	-40	-	+90	ºC

5.4. Power Consumption

5.4.1. EG91-AUX Power Consumption

Table 41: EG91-AUX Power Consumption

Description	Condition	Typ.	Unit
OFF state	Power down	10	μA

[20] To meet the normal operating temperature range requirements, it is necessary to ensure effective thermal dissipation, e.g., by adding passive or active heatsinks, heat pipes, vapor chambers. Within this range, the module's indicators comply with 3GPP specification requirements.

[21] To meet the extended operating temperature range requirements, it is necessary to ensure effective thermal dissipation, e.g., by adding passive or active heatsinks, heat pipes, vapor chambers. Within this range, the module remains the ability to establish and maintain functions such as voice, SMS and data transmission, without any unrecoverable malfunction. Radio spectrum and radio network remain uninfluenced, whereas the value of one or more parameters, such as P_{out}, may decrease and fall below the range of the 3GPP specified tolerances. When the temperature returns to the normal operating temperature range, the module's indicators will comply with 3GPP specification requirements again.

Sleep state	**AT+CFUN=0** (USB disconnected)	1.1	mA
	GSM850 DRX = 2 (USB disconnected)	2.0	mA
	GSM850 DRX = 5 (USB Suspend)	1.6	mA
	GSM850 DRX = 9 (USB disconnected)	1.4	mA
	WCDMA PF = 64 (USB disconnected)	1.8	mA
	WCDMA PF = 64 (USB Suspend)	2.1	mA
	WCDMA PF = 512 (USB disconnected)	1.2	mA
	LTE-FDD PF = 64 (USB disconnected)	2.3	mA
	LTE-FDD PF = 64 (USB Suspend)	2.8	mA
	LTE-FDD PF = 256 (USB disconnected)	1.5	mA
Idle state	GSM850 DRX = 5 (USB disconnected)	18	mA
	GSM850 DRX = 5 (USB connected)	28	mA
	WCDMA PF = 64 (USB disconnected)	17	mA
	WCDMA PF = 64 (USB connected)	28	mA
	LTE-FDD PF = 64 (USB disconnected)	23	mA
	LTE-FDD PF = 64 (USB connected)	34	mA
GPRS data transmission	GSM850 4DL/1UL @ 32.48 dBm	217.9	mA
	GSM850 3DL/2UL @ 31.89 dBm	372.3	mA
	GSM850 2DL/3UL @ 29.45 dBm	432.9	mA
	GSM850 1DL/4UL @ 28.31 dBm	513.9	mA
	EGSM900 4DL/1UL @ 33.17 dBm	235.1	mA
	EGSM900 3DL/2UL @ 32.16 dBm	387.7	mA
	EGSM900 2DL/3UL @ 29.77 dBm	446.5	mA
	EGSM900 1DL/4UL @ 28.59 dBm	540.0	mA
	DCS1800 4DL/1UL @ 30.19 dBm	154.4	mA

	DCS1800 3DL/2UL @ 29.23 dBm	258.0	mA
	DCS1800 2DL/3UL @ 27.19 dBm	332.4	mA
	DCS1800 1DL/4UL @ 26.14 dBm	419.1	mA
	PCS1900 4DL/1UL @ 30.22 dBm	155.0	mA
	PCS1900 3DL/2UL @ 29.48 dBm	259.5	mA
	PCS1900 2DL/3UL @ 27.50 dBm	333.1	mA
	PCS1900 1DL/4UL @ 26.44 dBm	416.8	mA
EDGE data transmission	GSM850 4DL/1UL PCL = 8 @ 25.75 dBm	161.8	mA
	GSM850 3DL/2UL PCL = 8 @ 25.49 dBm	291.8	mA
	GSM850 2DL/3UL PCL = 8 @ 23.26 dBm	410.2	mA
	GSM850 1DL/4UL PCL = 8 @ 22.01 dBm	520.5	mA
	EGSM900 4DL/1UL PCL = 8 @ 26.04 dBm	161.5	mA
	EGSM900 3DL/2UL PCL = 8 @ 25.86 dBm	294.6	mA
	EGSM900 2DL/3UL PCL = 8 @ 23.62 dBm	411.4	mA
	EGSM900 1DL/4UL PCL = 8 @ 22.27 dBm	520.8	mA
	DCS1800 4DL/1UL PCL = 2 @ 26.12 dBm	139.4	mA
	DCS1800 3DL/2UL PCL = 2 @ 25.02 dBm	250.7	mA
	DCS1800 2DL/3UL PCL = 2 @ 22.75 dBm	355.3	mA
	DCS1800 1DL/4UL PCL = 2 @ 21.47 dBm	452.1	mA
	PCS1900 4DL/1UL PCL = 2 @ 26.36 dBm	138.3	mA
	PCS1900 3DL/2UL PCL = 2 @ 25.2 dBm	248.2	mA
	PCS1900 2DL/3UL PCL = 2 @ 22.94 dBm	351.5	mA
	PCS1900 1DL/4UL PCL = 2 @ 21.67 dBm	448.8	mA
WCDMA data transmission	WCDMA B1 HSDPA @ 22.30 dBm	609.6	mA
	WCDMA B1 HSUPA @ 21.50 dBm	640.5	mA

	WCDMA B2 HSDPA @ 22.14 dBm	557.4	mA
	WCDMA B2 HSUPA @ 21.18 dBm	539.4	mA
	WCDMA B5 HSDPA @ 22.6 dBm	588.2	mA
	WCDMA B5 HSUPA @ 21.45 dBm	545.2	mA
	WCDMA B8 HSDPA @ 21.92 dBm	578.1	mA
	WCDMA B8 HSUPA @ 21.93 dBm	592.5	mA
LTE data transmission	LTE-FDD B1 @ 22.96 dBm	777.4	mA
	LTE-FDD B2 @ 22.79 dBm	634.4	mA
	LTE-FDD B3 @ 23.09 dBm	697.9	mA
	LTE-FDD B4 @ 22.83 dBm	704.6	mA
	LTE-FDD B5 @ 23.05 dBm	657.1	mA
	LTE-FDD B7 @ 22.71 dBm	765.3	mA
	LTE-FDD B8 @ 22.80 dBm	635.3	mA
	LTE-FDD B28 @ 22.84 dBm	670.0	mA
	LTE-FDD B66 @ 22.73 dBm	725.9	mA
GSM voice call	GSM850 PCL = 5 @ 32.57 dBm	227.8	mA
	EGSM900 PCL = 5 @ 33.21 dBm	253.8	mA
	DCS1800 PCL = 0 @ 30.24 dBm	168.0	mA
	PCS1900 PCL = 0 @ 30.33 dBm	166.8	mA
WCDMA voice call	WCDMA B1 @ 22.93 dBm	656.2	mA
	WCDMA B2 @ 22.95 dBm	579.8	mA
	WCDMA B5 @ 22.54 dBm	589.8	mA
	WCDMA B8 @ 22.47 dBm	627.8	mA

5.4.2. EG91-E Power Consumption

Table 42: EG91-E Power Consumption

Description	Condition	Typ.	Unit
OFF state	Power down	13	μA
Sleep state	**AT+CFUN=0** (USB disconnected)	1.1	mA
	EGSM900 DRX = 2 (USB disconnected)	2.0	mA
	EGSM900 DRX = 5 (USB Suspend)	1.9	mA
	EGSM900 DRX = 9 (USB disconnected)	1.3	mA
	WCDMA PF = 64 (USB disconnected)	1.7	mA
	WCDMA PF = 64 (USB Suspend)	2.1	mA
	WCDMA PF = 512 (USB disconnected)	1.1	mA
	LTE-FDD PF = 64 (USB disconnected)	2.1	mA
	LTE-FDD PF = 64 (USB Suspend)	2.6	mA
	LTE-FDD PF = 256 (USB disconnected)	1.4	mA
Idle state	EGSM900 DRX = 5 (USB disconnected)	19.0	mA
	EGSM900 DRX = 5 (USB connected)	29.0	mA
	WCDMA PF = 64 (USB disconnected)	19.0	mA
	WCDMA PF = 64 (USB connected)	29.0	mA
	LTE-FDD PF = 64 (USB disconnected)	19.0	mA
	LTE-FDD PF = 64 (USB connected)	29.0	mA
GPRS data transmission	EGSM900 4DL/1UL @ 32.67 dBm	260	mA
	EGSM900 3DL/2UL @ 32.59 dBm	463	mA
	EGSM900 2DL/3UL @ 30.74 dBm	552	mA
	EGSM900 1DL/4UL @ 29.26 dBm	619	mA
	DCS1800 4DL/1UL @ 29.2 dBm	165	mA

	DCS1800 3DL/2UL @ 29.13 dBm	267	mA
	DCS1800 2DL/3UL @ 29.01 dBm	406	mA
	DCS1800 1DL/4UL @ 28.86 dBm	467	mA
EDGE data transmission	EGSM900 4DL/1UL PCL = 8 @ 27.1 dBm	163	mA
	EGSM900 3DL/2UL PCL = 8 @ 27.16 dBm	274	mA
	EGSM900 2DL/3UL PCL = 8 @ 26.91 dBm	383	mA
	EGSM900 1DL/4UL PCL = 8 @ 26.12 dBm	463	mA
	DCS1800 4DL/1UL PCL = 2 @ 25.54 dBm	136	mA
	DCS1800 3DL/2UL PCL = 2 @ 25.68 dBm	220	mA
	DCS1800 2DL/3UL PCL = 2 @ 25.61 dBm	306	mA
	DCS1800 1DL/4UL PCL = 2 @ 25.41 dBm	396	mA
WCDMA data transmission	WCDMA B1 HSDPA @ 22.29 dBm	507	mA
	WCDMA B1 HSUPA @ 21.79 dBm	516	mA
	WCDMA B8 HSDPA @ 22.47 dBm	489	mA
	WCDMA B8 HSUPA @ 21.98 dBm	482	mA
LTE data transmission	LTE-FDD B1 @ 22.98 dBm	685	mA
	LTE-FDD B3 @ 23.23 dBm	698	mA
	LTE-FDD B7 @ 23.46 dBm	723	mA
	LTE-FDD B8 @ 23.35 dBm	655	mA
	LTE-FDD B20 @ 23.41 dBm	723	mA
	LTE-FDD B28A @ 23.16 dBm	660	mA
GSM voice call	EGSM900 PCL = 5 @ 32.5 dBm	258	mA
	DCS1800 PCL = 0 @ 29.23 dBm	159	mA
WCDMA voice call	WCDMA B1 @ 23.06 dBm	555	mA
	WCDMA B8 @ 23.45 dBm	535	mA

5.4.3. EG91-EX Power Consumption

Table 43: EG91-EX Power Consumption

Description	Condition	Typ.	Unit
OFF state	Power down	15	μA
Sleep state	**AT+CFUN=0** (USB disconnected)	1.3	mA
	EGSM900 DRX = 2 (USB disconnected)	2.3	mA
	EGSM900 DRX = 5 (USB Suspend)	2.0	mA
	EGSM900 DRX = 9 (USB disconnected)	1.6	mA
	WCDMA PF = 64 (USB disconnected)	1.8	mA
	WCDMA PF = 64 (USB Suspend)	2.1	mA
	WCDMA PF = 512 (USB disconnected)	1.3	mA
	LTE-FDD PF = 64 (USB disconnected)	2.3	mA
	LTE-FDD PF = 64 (USB Suspend)	2.6	mA
	LTE-FDD PF = 256 (USB disconnected)	1.5	mA
Idle state	EGSM900 DRX = 5 (USB disconnected)	21.0	mA
	EGSM900 DRX = 5 (USB connected)	31.0	mA
	WCDMA PF = 64 (USB disconnected)	21.0	mA
	WCDMA PF = 64 (USB connected)	31.0	mA
	LTE-FDD PF = 64 (USB disconnected)	21.0	mA
	LTE-FDD PF = 64 (USB connected)	31.0	mA
GPRS data transmission	EGSM900 4DL/1UL @ 33.06 dBm	247.9	mA
	EGSM900 3DL/2UL @ 32.93 dBm	450.8	mA
	EGSM900 2DL/3UL @ 31.1 dBm	536.4	mA
	EGSM900 1DL/4UL @ 29.78 dBm	618	mA
	DCS1800 4DL/1UL @ 29.3 dBm	144	mA

	DCS1800 3DL/2UL @ 29.3 dBm	253.4	mA
	DCS1800 2DL/3UL @ 29.21 dBm	355.4	mA
	DCS1800 1DL/4UL @ 29.07 dBm	455.7	mA
EDGE data transmission	EGSM900 4DL/1UL PCL = 8 @ 27.29 dBm	169.5	mA
	EGSM900 3DL/2UL PCL = 8 @ 27.01 dBm	305.06	mA
	EGSM900 2DL/3UL PCL = 8 @ 26.86 dBm	434	mA
	EGSM900 1DL/4UL PCL = 8 @ 25.95 dBm	548	mA
	DCS1800 4DL/1UL PCL = 2 @ 26.11 dBm	135	mA
	DCS1800 3DL/2UL PCL = 2 @ 25.8 dBm	244	mA
	DCS1800 2DL/3UL PCL = 2 @ 25.7 dBm	349	mA
	DCS1800 1DL/4UL PCL = 2 @ 25.6 dBm	455	mA
WCDMA data transmission	WCDMA B1 HSDPA @ 22.48 dBm	485	mA
	WCDMA B1 HSUPA @ 21.9 dBm	458	mA
	WCDMA B8 HSDPA @ 22.6 dBm	556	mA
	WCDMA B8 HSUPA @ 22.02 dBm	520	mA
LTE data transmission	LTE-FDD B1 @ 23.37 dBm	605	mA
	LTE-FDD B3 @ 23.3 dBm	667	mA
	LTE-FDD B7 @ 23.2 dBm	783	mA
	LTE-FDD B8 @ 23.09 dBm	637	mA
	LTE-FDD B20 @ 23.21 dBm	646	mA
	LTE-FDD B28 @ 22.76 dBm	661	mA
GSM voice call	EGSM900 PCL = 5 @ 32.36 dBm	259	mA
	DCS1800 PCL = 0 @ 29.5 dBm	149	mA
WCDMA voice call	WCDMA B1 @ 23.4 dBm	494	mA
	WCDMA B8 @ 23.6 dBm	608	mA

5.4.4. EG91-JP Power Consumption

Table 44: EG91-JP Power Consumption

Description	Condition	Typ.	Unit
OFF state	Power down	9.3	μA
Sleep state	**AT+CFUN=0** (USB disconnected)	0.8	mA
	AT+CFUN=0 (USB Suspend)	1	mA
	AT+CFUN=4 (USB disconnected)	0.9	mA
	AT+CFUN=4 (USB Suspend)	1.1	mA
	LTE-FDD PF = 32 (USB disconnected)	2.9	mA
	LTE-FDD PF = 64 (USB disconnected)	1.9	mA
	LTE-FDD PF = 64 (USB Suspend)	2.2	mA
	LTE-FDD PF = 128 (USB disconnected)	1.4	mA
	LTE-FDD PF = 256 (USB disconnected)	1.2	mA
Idle state	LTE-FDD PF = 64 (USB disconnected)	14.6	mA
	LTE-FDD PF = 64 (USB Suspend)	24.3	mA
LTE data transmission	LTE-FDD B1 @ 23.32 dBm	760	mA
	LTE-FDD B3 @ 23.10 dBm	770	mA
	LTE-FDD B8 @ 23.51 dBm	660	mA
	LTE-FDD B18 @ 24.01 dBm	700	mA
	LTE-FDD B19 @ 24.01 dBm	700	mA
	LTE-FDD B26 @ 23.50 dBm	700	mA

5.4.5. EG91-NA Power Consumption

Table 45: EG91-NA Power Consumption

Description	Condition	Typ.	Unit
OFF state	Power down	13	μA
Sleep state	**AT+CFUN=0** (USB disconnected)	1.0	mA
	WCDMA PF = 64 (USB disconnected)	2.2	mA
	WCDMA PF = 64 (USB Suspend)	2.5	mA
	WCDMA PF = 512 (USB disconnected)	1.4	mA
	LTE-FDD PF = 64 (USB disconnected)	2.6	mA
	LTE-FDD PF = 64 (USB Suspend)	2.9	mA
	LTE-FDD PF = 256 (USB disconnected)	1.7	mA
Idle state	WCDMA PF = 64 (USB disconnected)	14.0	mA
	WCDMA PF = 64 (USB connected)	26.0	mA
	LTE-FDD PF = 64 (USB disconnected)	15.0	mA
	LTE-FDD PF = 64 (USB connected)	26.0	mA
WCDMA data transmission	WCDMA B2 HSDPA @ 22.45 dBm	569	mA
	WCDMA B2 HSUPA @ 21.73 dBm	559	mA
	WCDMA B4 HSDPA @ 23.05 dBm	572	mA
	WCDMA B4 HSUPA @ 22.86 dBm	586	mA
	WCDMA B5 HSDPA @ 23 dBm	518	mA
	WCDMA B5 HSUPA @ 22.88 dBm	514	mA
LTE data transmission	LTE-FDD B2 @ 23.29 dBm	705	mA
	LTE-FDD B4 @ 23.19 dBm	693	mA
	LTE-FDD B5 @ 23.39 dBm	601	mA
	LTE-FDD B12 @ 23.16 dBm	650	mA

	LTE-FDD B13 @ 23.36 dBm	602	mA
WCDMA voice call	WCDMA B2 @ 23.34 dBm	627	mA
	WCDMA B4 @ 23.47 dBm	591	mA
	WCDMA B5 @ 23.37 dBm	536	mA

5.4.6. EG91-NAX Power Consumption

Table 46: EG91-NAX Power Consumption

Description	Condition	Typ.	Unit
OFF state	Power down	9	μA
Sleep state	**AT+CFUN=0** (USB disconnected)	1.1	mA
	WCDMA PF = 64 (USB disconnected)	2.1	mA
	WCDMA PF = 64 (USB Suspend)	2.2	mA
	WCDMA PF = 512 (USB disconnected)	1.6	mA
	LTE-FDD PF = 64 (USB disconnected)	2.6	mA
	LTE-FDD PF = 64 (USB Suspend)	2.7	mA
	LTE-FDD PF = 256 (USB disconnected)	1.8	mA
Idle state	WCDMA PF = 64 (USB disconnected)	16.7	mA
	WCDMA PF = 64 (USB connected)	32.2	mA
	LTE-FDD PF = 64 (USB disconnected)	14.0	mA
	LTE-FDD PF = 64 (USB connected)	32.6	mA
WCDMA data transmission	WCDMA B2 HSDPA @ 21.74 dBm	528	mA
	WCDMA B2 HSUPA @ 21.47 dBm	536	mA
	WCDMA B4 HSDPA @ 22.67 dBm	542	mA
	WCDMA B4 HSUPA @ 22.30 dBm	550	mA
	WCDMA B5 HSDPA @ 22.63 dBm	523	mA

	WCDMA B5 HSUPA @ 22.31 dBm	523	mA
LTE data transmission	LTE-FDD B2 @ 23.08 dBm	694	mA
	LTE-FDD B4 @ 23.31 dBm	691	mA
	LTE-FDD B5 @ 23.23 dBm	586	mA
	LTE-FDD B12 @ 23.03 dBm	613	mA
	LTE-FDD B13 @ 23.13 dBm	626	mA
	LTE-FDD B25 @ 22.96 dBm	689	mA
	LTE-FDD B26 @ 23.11 dBm	636	mA
WCDMA voice call	WCDMA B2 @ 23.08 dBm	581	mA
	WCDMA B4 @ 23.21 dBm	557	mA
	WCDMA B5 @ 23.29 dBm	534	mA

5.4.7. EG91-NAXD Power Consumption

Table 47: EG91-NAXD Power Consumption

Description	Condition	Typ.	Unit
OFF state	Power down	9	μA
Sleep state	**AT+CFUN=0** (USB disconnected)	1.1	mA
	WCDMA PF = 64 (USB disconnected)	2.1	mA
	WCDMA PF = 64 (USB Suspend)	2.2	mA
	WCDMA PF = 512 (USB disconnected)	1.6	mA
	LTE-FDD PF = 64 (USB disconnected)	2.6	mA
	LTE-FDD PF = 64 (USB Suspend)	2.7	mA
	LTE-FDD PF = 256 (USB disconnected)	1.8	mA
Idle state	WCDMA PF = 64 (USB disconnected)	16.7	mA
	WCDMA PF = 64 (USB connected)	32.2	mA

	LTE-FDD PF = 64 (USB disconnected)	14.0	mA
	LTE-FDD PF = 64 (USB connected)	32.6	mA
WCDMA data transmission	WCDMA B2 HSDPA @ 21.74 dBm	528	mA
	WCDMA B2 HSUPA @ 21.47 dBm	536	mA
	WCDMA B4 HSDPA @ 22.67 dBm	542	mA
	WCDMA B4 HSUPA @ 22.30 dBm	550	mA
	WCDMA B5 HSDPA @ 22.63 dBm	523	mA
	WCDMA B5 HSUPA @ 22.31 dBm	523	mA
LTE data transmission	LTE-FDD B2 @ 23.08 dBm	694	mA
	LTE-FDD B4 @ 23.31 dBm	691	mA
	LTE-FDD B5 @ 23.23 dBm	586	mA
	LTE-FDD B12 @ 23.03 dBm	613	mA
	LTE-FDD B13 @ 23.13 dBm	626	mA
	LTE-FDD B25 @ 22.96 dBm	689	mA
	LTE-FDD B26 @ 23.11 dBm	636	mA

5.4.8. EG91-VX Power Consumption

Table 48: EG91-VX Power Consumption

Description	Condition	Typ.	Unit
OFF state	Power down	7.6	μA
Sleep state	**AT+CFUN=0** (USB disconnected)	0.95	mA
	AT+CFUN=0 (USB Suspend)	1.19	mA
	AT+CFUN=4 (USB disconnected)	0.98	mA
	AT+CFUN=4 (USB Suspend)	1.25	mA
	LTE-FDD PF = 32 (USB disconnected)	3.44	mA

	LTE-FDD PF = 64 (USB disconnected)	2.41	mA
	LTE-FDD PF = 64 (USB Suspend)	2.65	mA
	LTE-FDD PF = 128 (USB disconnected)	1.88	mA
	LTE-FDD PF = 256 (USB disconnected)	1.6	mA
Idle state	LTE-FDD PF = 64 (USB disconnected)	16.5	mA
	LTE-FDD PF = 64 (USB connected)	30.8	mA
LTE data transmission	LTE-FDD B4 @ 23.36 dBm	715	mA
	LTE-FDD B13 @ 23.38 dBm	642	mA

NOTE

The above power consumption data are tested under 50 Ω impedance.

5.4.9. GNSS Power Consumption

Table 49: GNSS Power Consumption

Description	Condition	Typ.	Unit
Acquisition (**AT+CFUN=0**)	Cold start @ Passive antenna	54	mA
	Hot start @ Passive antenna	54	mA
	Lost state @ Passive antenna	53	mA
Tracking (**AT+CFUN=0**)	Open sky @ Passive antenna	32	mA

5.5. ESD Protection

Static electricity occurs naturally and it may damage the module. Therefore, applying proper ESD countermeasures and handling methods is imperative. For example, wear anti-static gloves during the development, production, assembly and testing of the module; add ESD protection components to the ESD sensitive interfaces and points in the product design.

Table 50: Electrostatic Discharge Characteristics (Temperature: 25–30 ºC, Humidity: 40 ±5 %)

Tested Interface	Contact Discharge	Air Discharge	Unit
VBAT, GND	±5	±10	kV
All Antenna Interfaces	±4	±8	kV
Other Interfaces	±0.5	±1	kV

5.6. Thermal Dissipation

In order to achieve better performance of the module, it is recommended to comply with the following principles for thermal consideration:

- On your PCB design, please keep placement of the module away from heating sources, especially high power components such as ARM processor, audio power amplifier and power supply.
- Do not place components on the opposite side of the PCB area where the module is mounted, in order to facilitate adding of heatsink when necessary.
- Do not apply solder mask on the opposite side of the PCB area where the module is mounted, so as to ensure better heat dissipation performance.
- The reference ground of the area where the module is mounted should be complete, and add ground vias as many as possible for better heat dissipation.
- Make sure the ground pads of the module and PCB are fully connected.
- According to your application demands, the heatsink can be mounted on the top of the module, or the opposite side of the PCB area where the module is mounted, or both of them.
- The heatsink should be designed with as many fins as possible to increase heat dissipation area. Meanwhile, a thermal pad with high thermal conductivity should be used between the heatsink and module/PCB.

The following shows two kinds of heatsink designs for reference and you can choose one or both of them according to their application structure.

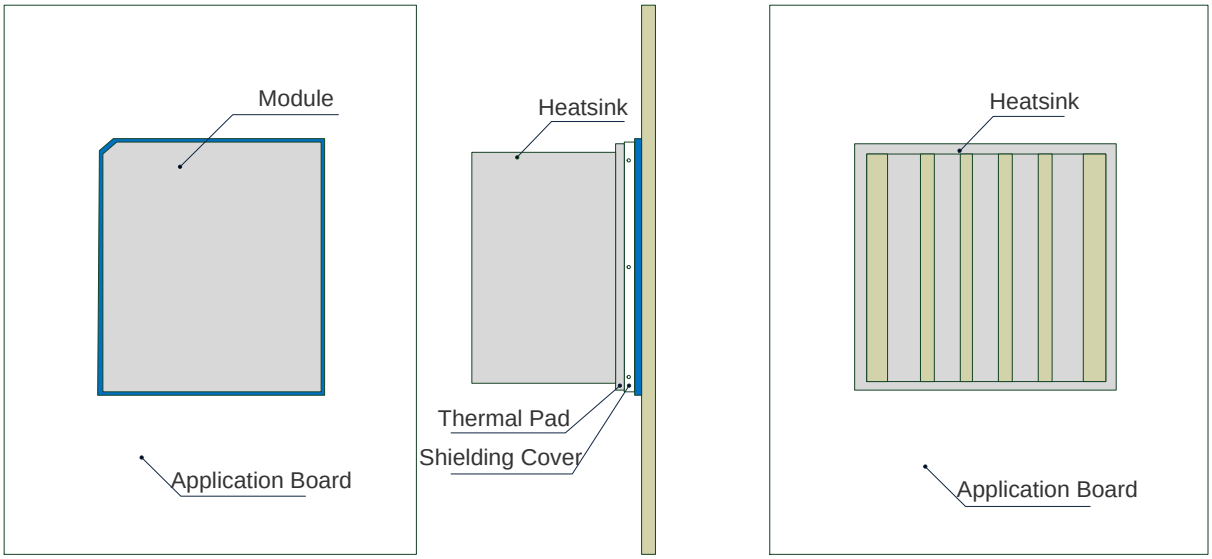


Figure 40: Referenced Heatsink Design (Heatsink at the Top of the Module)

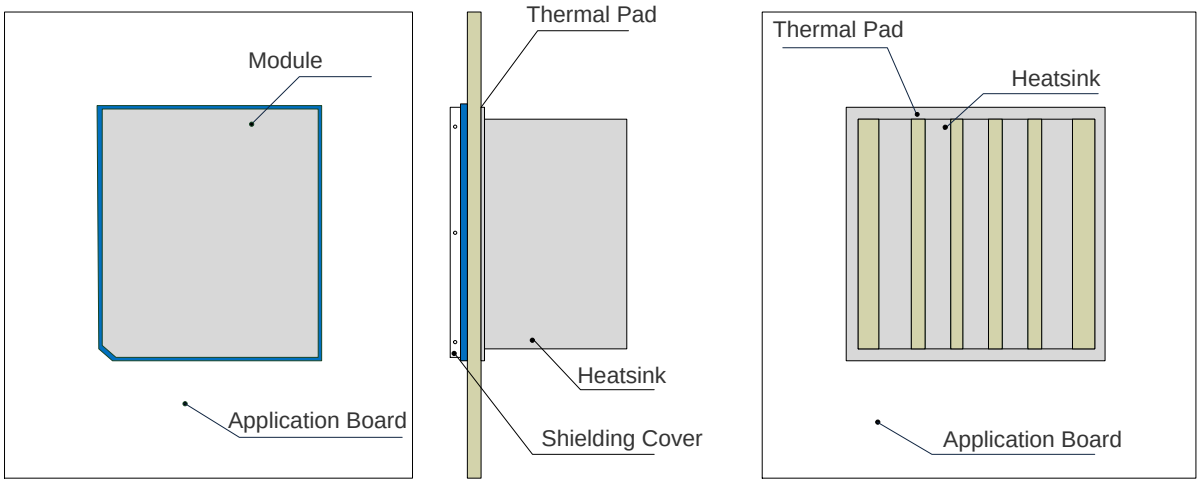


Figure 41: Referenced Heatsink Design (Heatsink at the Backside of Customers' PCB)

NOTE

The module offers the best performance when the internal BB chip stays below 105 °C. When the maximum temperature of the BB chip reaches or exceeds 105 °C, the module works normal but provides reduced performance (such as RF output power and data rate). When the maximum BB chip temperature reaches or exceeds 115 °C, the module will disconnect from the network, and it will recover to network connected state after the maximum temperature falls below 115 °C. Therefore, the thermal design should be maximally optimized to make sure the maximum BB chip temperature always maintains below 105 °C. You can execute **AT+QTEMP** and get the maximum BB chip temperature from the first returned value. For more details, see ***document [8]***.

6 Mechanical Information

This chapter describes the mechanical dimensions of the module. All dimensions are measured in millimeter (mm), and the dimensional tolerances are ±0.2 mm unless otherwise specified.

6.1. Mechanical Dimensions

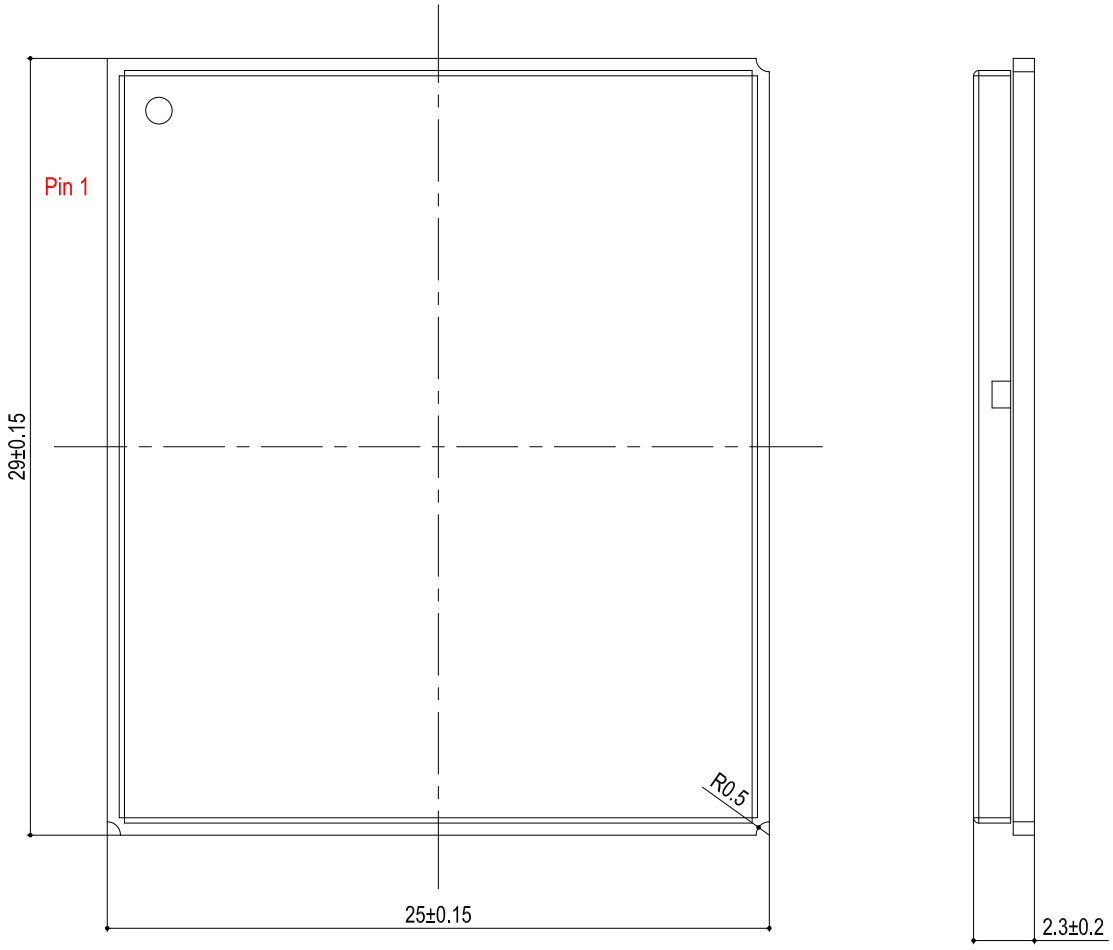


Figure 42: EG91-E Top and Side Dimensions

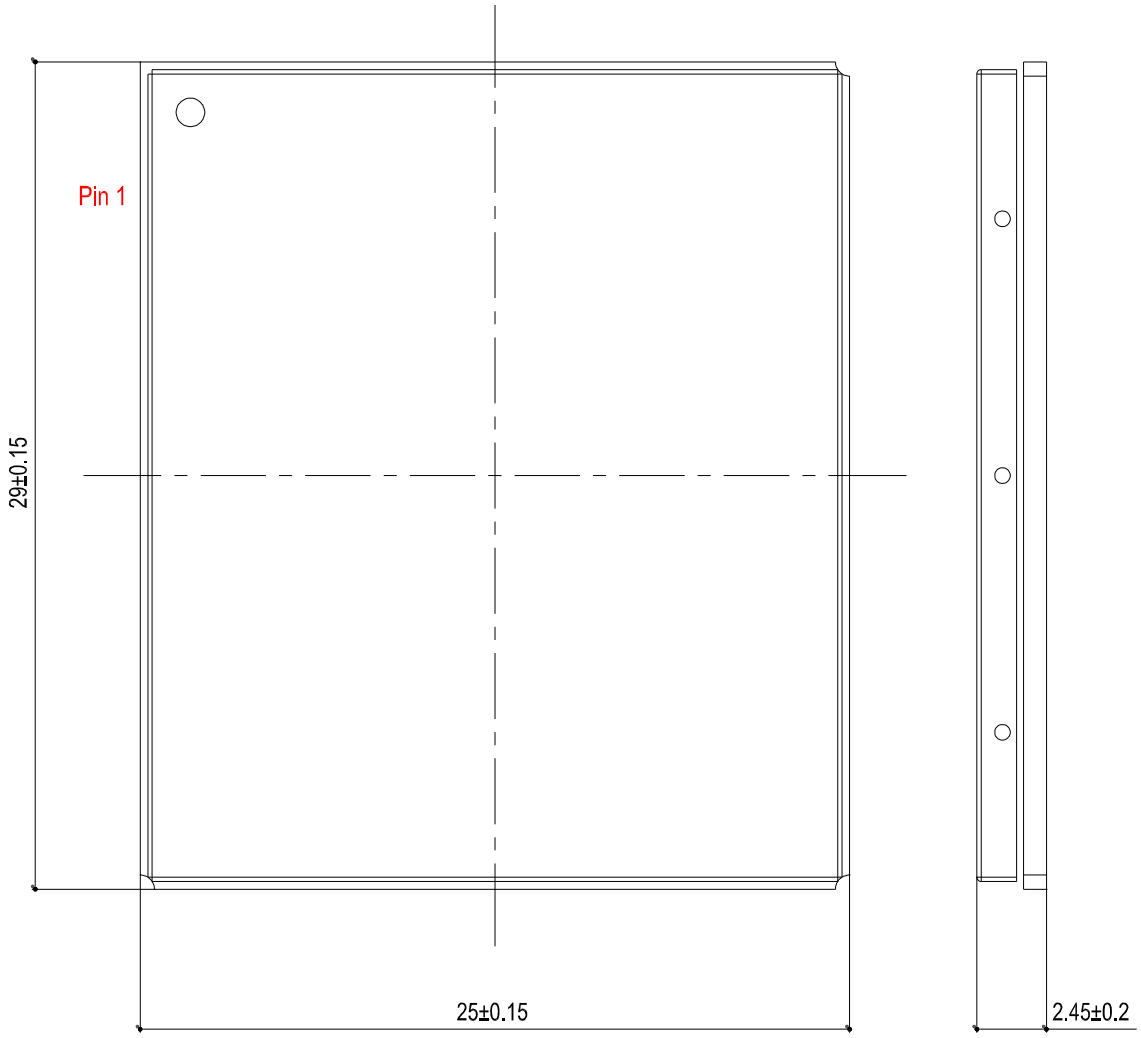


Figure 43: EG91-AUX/-EX/-JP/-VX Top and Side Dimensions

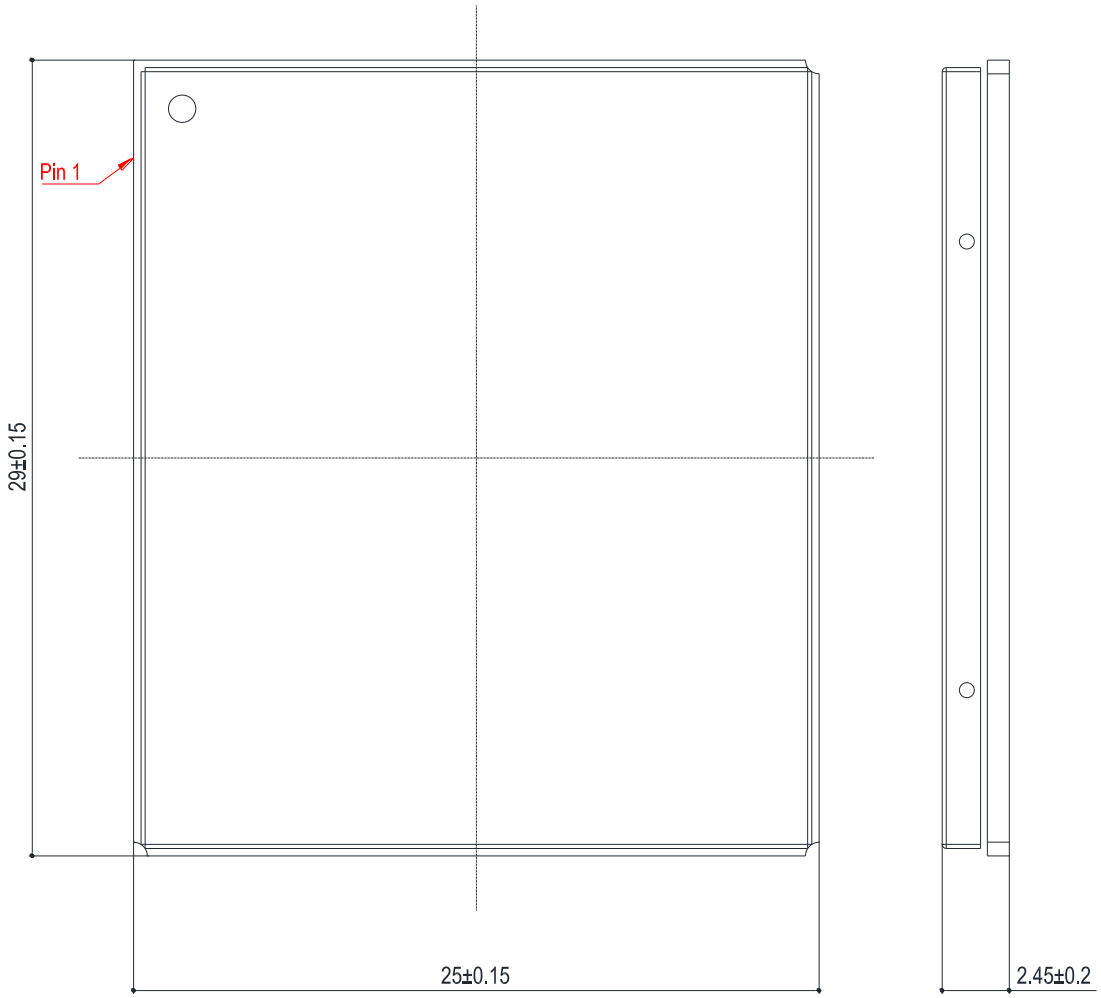


Figure 44: EG91-NA/-NAX/-NAXD Top and Side Dimensions

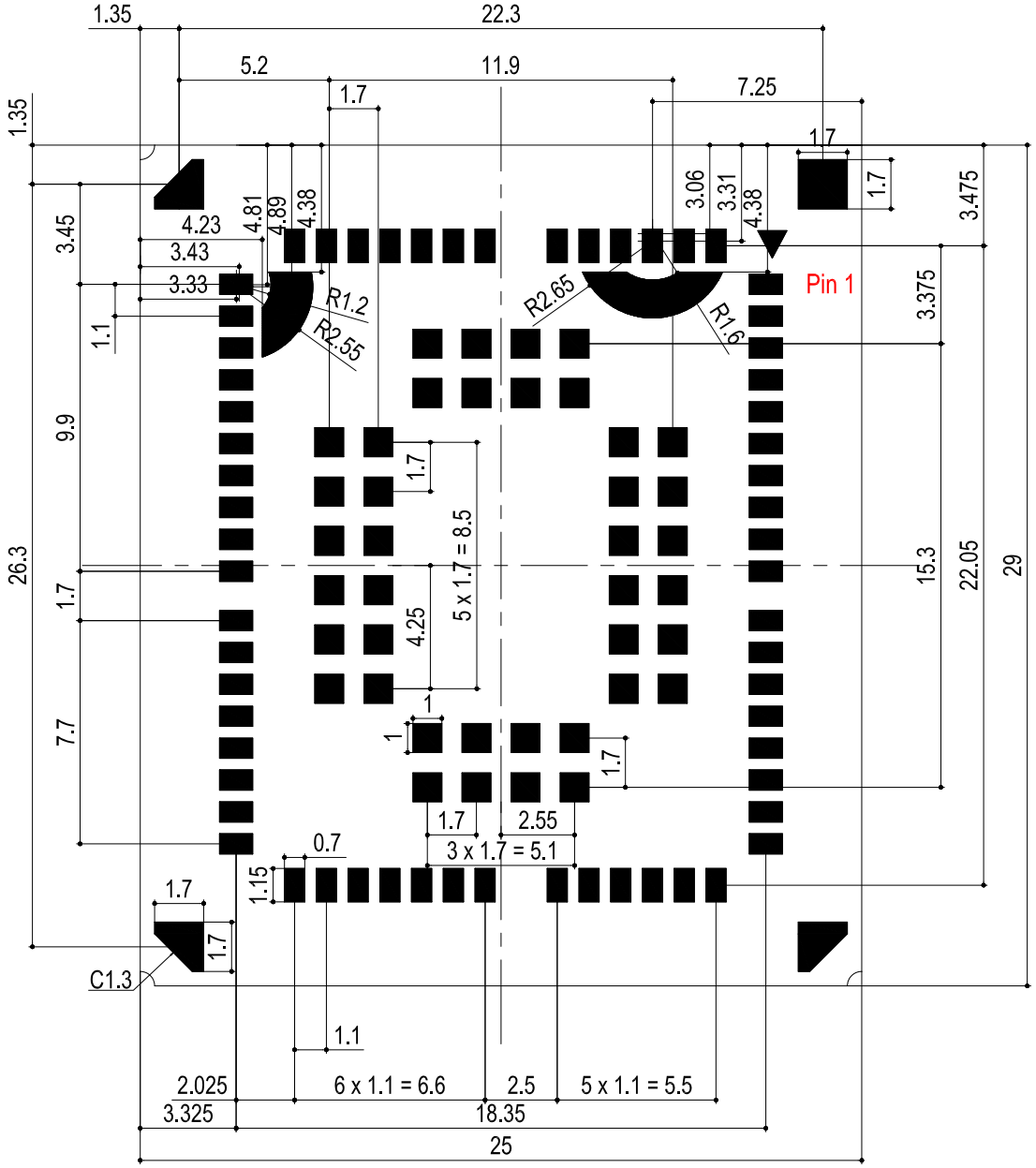


Figure 45: EG91-E Bottom Dimensions (Bottom View)

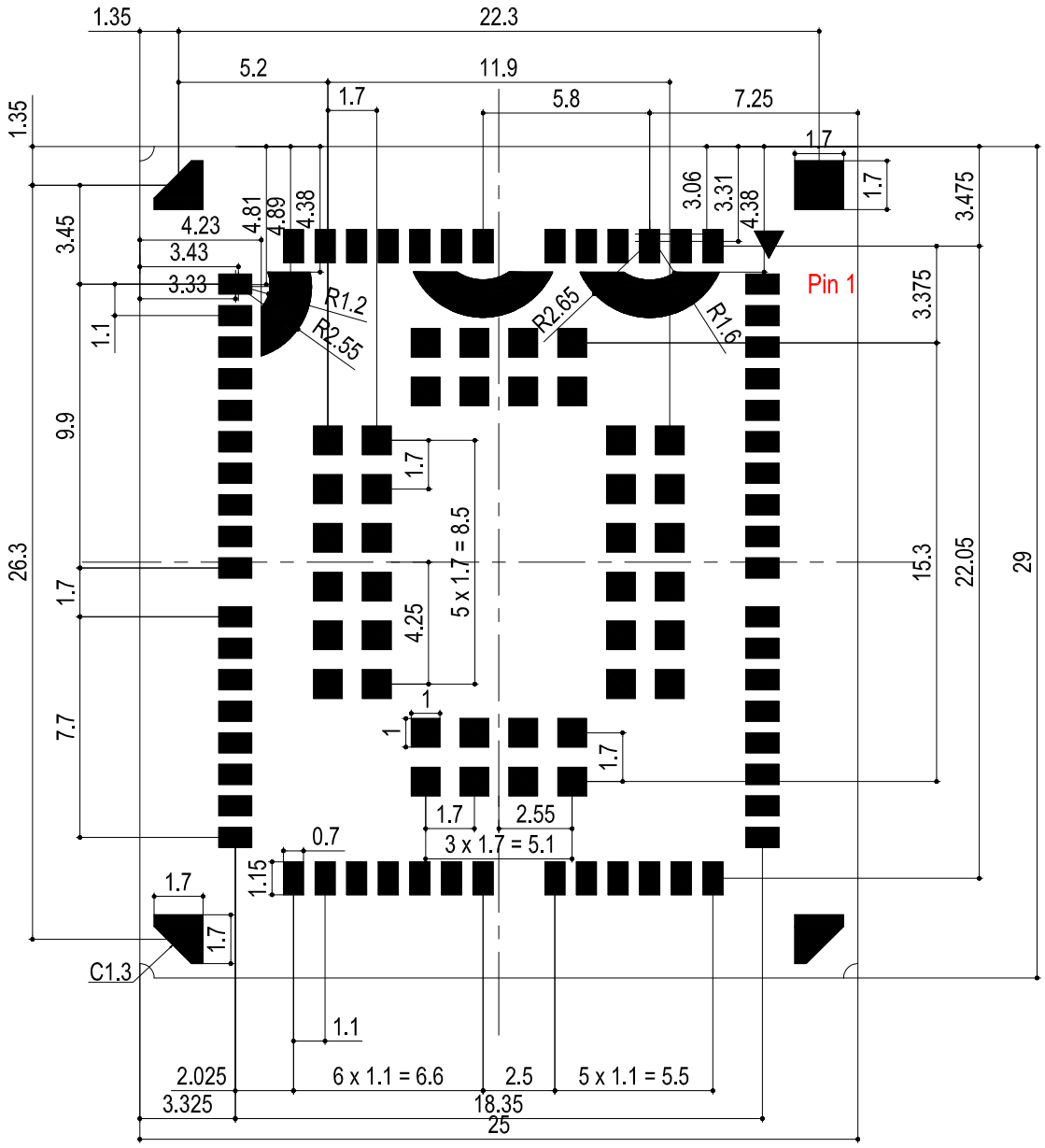


Figure 46: EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX Bottom Dimensions (Bottom View)

NOTE

The module's coplanarity standard: ≤ 0.13 mm.

6.2. Recommended Footprint

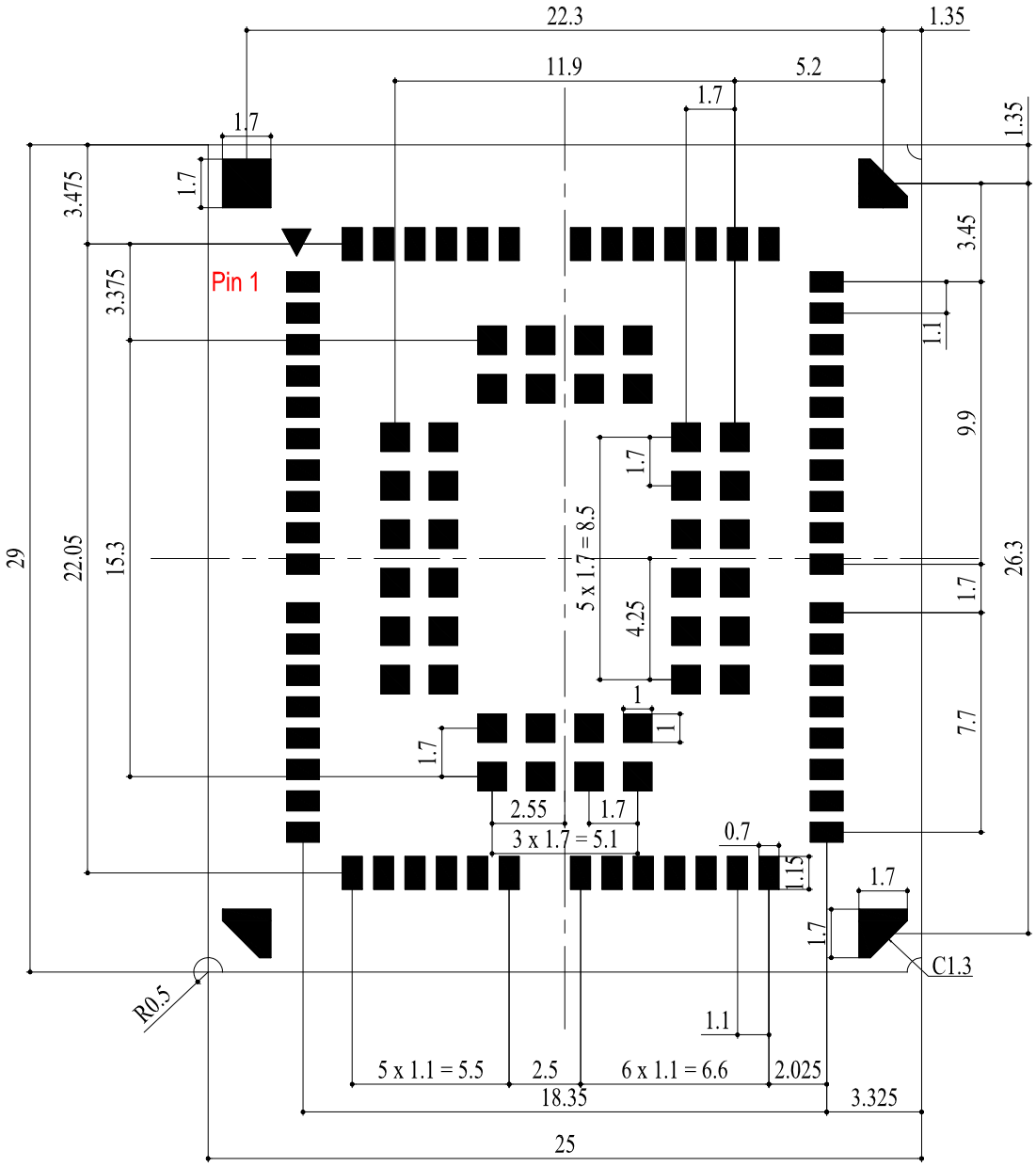
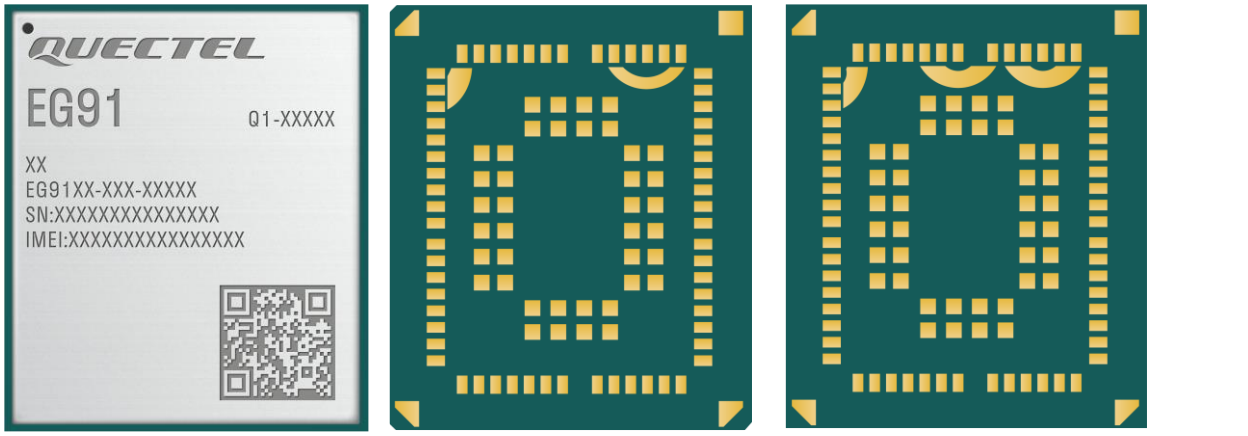


Figure 47: Recommended Footprint

NOTE

Keep at least 3 mm between the module and other components on the motherboard to improve soldering quality and maintenance convenience.

6.3. Top and Bottom Views



Top View

EG91-E Bottom View

EG91-AUX/-EX/-JP/-NA/-NAX/-NAXD/-VX Bottom View

Figure 48: Top and Bottom Views of the Module

NOTE

Images above are for illustration purpose only and may differ from the actual module. For authentic appearance and label, please refer to the module received from Quectel.

7 Storage, Manufacturing and Packaging

7.1. Storage Conditions

The module is provided with vacuum-sealed packaging. MSL of the module is rated as 3. The storage requirements are shown below.

1. Recommended Storage Condition: the temperature should be 23 ±5 °C and the relative humidity should be 35–60 %.

2. Shelf life (in a vacuum-sealed packaging): 12 months in Recommended Storage Condition.

3. Floor life: 168 hours [22] in a factory where the temperature is 23 ±5 °C and relative humidity is below 60 %. After the vacuum-sealed packaging is removed, the module must be processed in reflow soldering or other high-temperature operations within 168 hours. Otherwise, the module should be stored in an environment where the relative humidity is less than 10 % (e.g., a dry cabinet).

4. The module should be pre-baked to avoid blistering, cracks and inner-layer separation in PCB under the following circumstances:

   - The module is not stored in Recommended Storage Condition;
   - Violation of the third requirement mentioned above;
   - Vacuum-sealed packaging is broken, or the packaging has been removed for over 24 hours;
   - Before module repairing.

5. If needed, the pre-baking should follow the requirements below:

   - The module should be baked for 24 hours at 120 ±5 °C;
   - The module must be soldered to PCB within 24 hours after the baking, otherwise it should be put in a dry environment such as in a dry cabinet.

[22] This floor life is only applicable when the environment conforms to *IPC/JEDEC J-STD-033*. It is recommended to start the solder reflow process within 24 hours after the package is removed if the temperature and moisture do not conform to, or are not sure to conform to *IPC/JEDEC J-STD-033*. And do not unpack the modules in large quantities until they are ready for soldering.

NOTE

1. To avoid blistering, layer separation and other soldering issues, extended exposure of the module to the air is forbidden.
2. Take out the module from the package and put it on high-temperature-resistant fixtures before baking. If shorter baking time is desired, see *IPC/JEDEC J-STD-033* for the baking procedure.
3. Pay attention to ESD protection, such as wearing anti-static gloves, when touching the modules.

7.2. Manufacturing and Soldering

Push the squeegee to apply the solder paste on the surface of stencil, thus making the paste fill the stencil openings and then penetrate to the PCB. Apply proper force on the squeegee to produce a clean stencil surface on a single pass. To guarantee module soldering quality, the thickness of stencil for the module is recommended to be 0.13–0.18 mm. For more details, see ***document [9]***.

The recommended peak reflow temperature should be 235–246 ºC, with 246 ºC as the absolute maximum reflow temperature. To avoid damage to the module caused by repeated heating, it is recommended that the module should be mounted only after reflow soldering for the other side of PCB has been completed. The recommended reflow soldering thermal profile (lead-free reflow soldering) and related parameters are shown below.

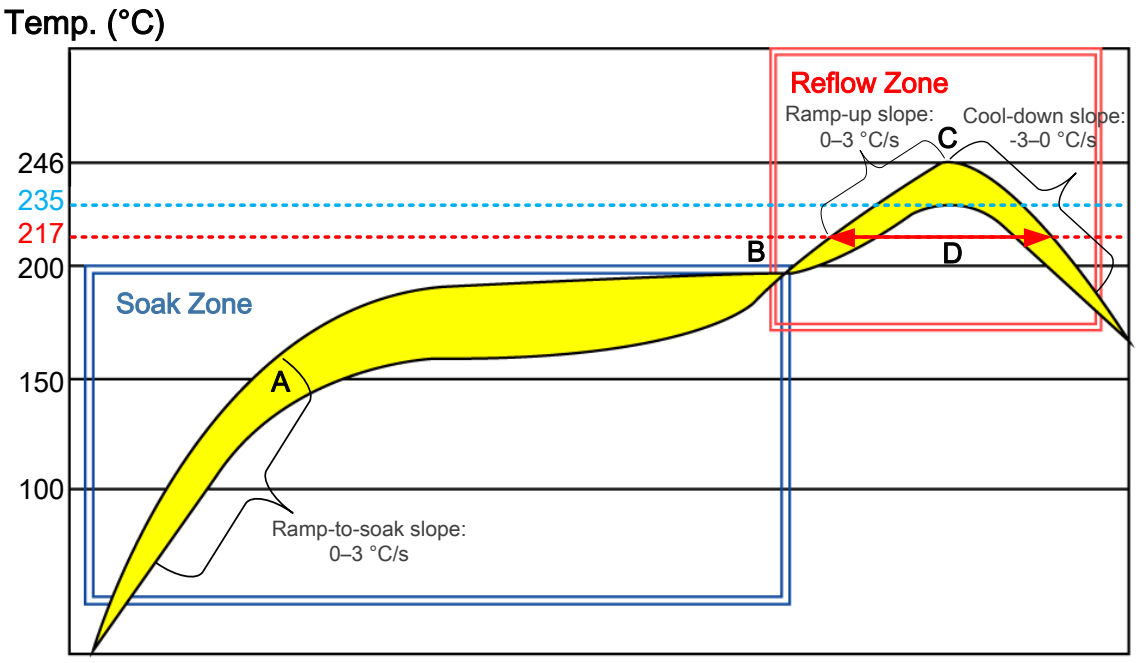


Figure 49: Recommended Reflow Soldering Thermal Profile

Table 51: Recommended Thermal Profile Parameters

Factor	Recommended Value
Soak Zone	
Ramp-to-soak slope	0–3 °C/s
Soak time (between A and B: 150 °C and 200 °C)	70–120 s
Reflow Zone	
Ramp-up slope	0–3 °C/s
Reflow time (D: over 217°C)	40–70 s
Max. temperature	235–246 °C
Cool-down slope	-3–0 °C/s
Reflow Cycle	
Max. reflow cycle	1

NOTE

1. The above profile parameter requirements are for the measured temperature of the solder joints. Both the hottest and coldest spots of solder joints on the PCB should meet the above requirements.
2. **EG91-E:**
   1) During manufacturing and soldering, or any other processes that may contact the module directly, NEVER wipe the module's shielding can with organic solvents, such as acetone, ethyl alcohol, isopropyl alcohol and trichloroethylene. Otherwise, the shielding can may become rusted.
   2) The shielding can for the module is made of Cupro-Nickel base material. It is tested that after 12 hours' Neutral Salt Spray test, the laser engraved label information on the shielding can is still clearly identifiable and the QR code is still readable, although white rust may be found.
3. **EG91 Series (EG91-AUX/-E/-EX/-JP/-NA/-NAX/-NAXD/-VX):**
   1) If a conformal coating is necessary for the module, do NOT use any coating material that may chemically react with the PCB or shielding cover, and prevent the coating material from flowing into the module.
   2) Avoid using ultrasonic technology for module cleaning since it can damage crystals inside the module.
   3) Avoid using materials that contain mercury (Hg), such as adhesives, for module processing, even if the materials are RoHS compliant and their mercury content is below 1000 ppm (0.1 %).
   4) Corrosive gases may corrode the electronic components inside the module, affecting their

reliability and performance, and potentially leading to a shortened service life that fails to meet the designed lifespan. Therefore, do not store or use unprotected modules in environments containing corrosive gases such as hydrogen sulfide, sulfur dioxide, chlorine, and ammonia.

5) Due to the complexity of the SMT process, please contact Quectel Technical Support in advance for any situation that you are not sure about, or any process (e.g. selective soldering, ultrasonic soldering) that is not mentioned in ***document [10]***.

7.3. Packaging Specification

This chapter outlines the key packaging parameters and processes. All figures below are for reference purposes only, as the actual appearance and structure of packaging materials may vary in delivery.

The modules are packed in a tape and reel packaging as specified in the sub-chapters below.

7.3.1. Carrier Tape

Carrier tape dimensions are illustrated in the following figure and table.

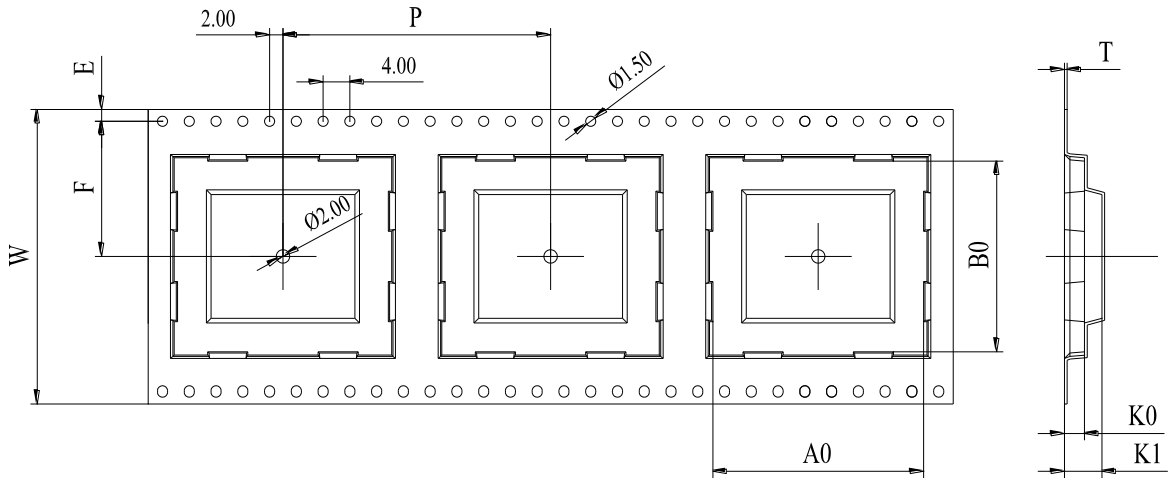


Figure 50: Carrier Tape Dimension Drawing (Unit: mm)

Table 52: Carrier Tape Dimension Table (Unit: mm)

W	P	T	A0	B0	K0	K1	F	E
44	32	0.35	25.5	29.5	3.2	5.8	20.2	1.75

NOTE

Actual pads in kind prevail when mounting.

7.3.2. Plastic Reel

Plastic reel dimensions are illustrated in the following figure and table:

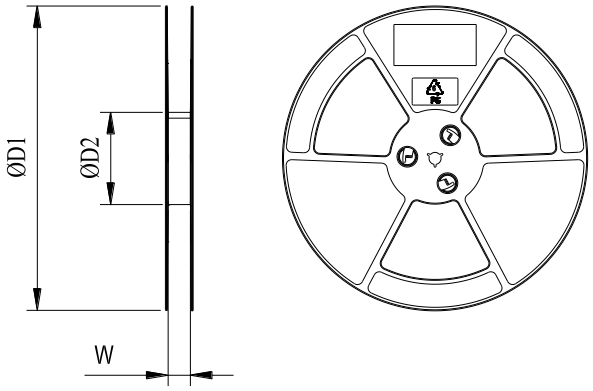


Figure 51: Plastic Reel Dimension Drawing

Table 53: Plastic Reel Dimension Table (Unit: mm)

øD1	øD2	W
330	100	44.5

7.3.3. Mounting Direction

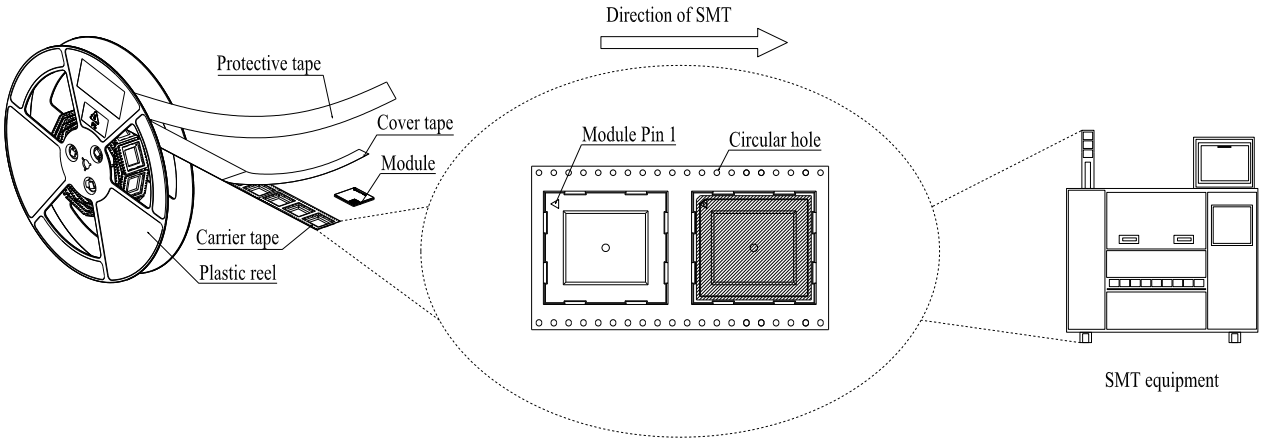
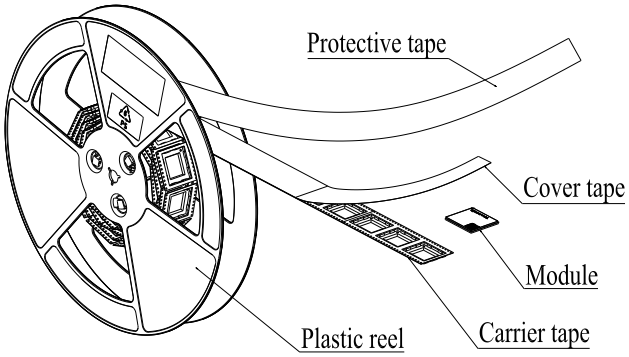


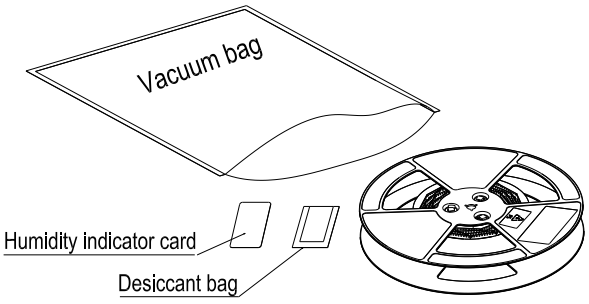
Figure 52: Mounting Direction

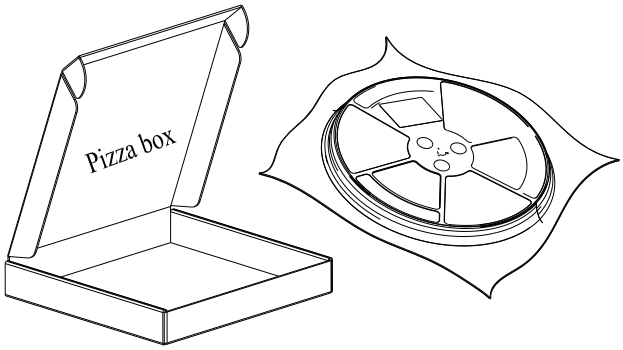
7.3.4. Packaging Process



Place the modules onto the carrier tape cavity and cover them securely with cover tape. Wind the heat-sealed carrier tape onto a plastic reel and apply a protective tape for additional protection. 1 plastic reel can load 250 modules.

Place the packaged plastic reel, humidity indicator card and desiccant bag into a vacuum bag, and vacuumize it.





Place the vacuum-packed plastic reel into a pizza box.

Place the 4 packaged pizza boxes into 1 carton and seal it. 1 carton can pack 1000 modules.

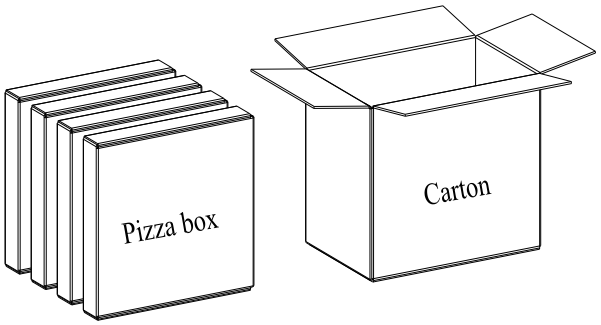


Figure 53: Packaging Process

8 Appendix References

Table 54: Related Documents

Document Name
[1] Quectel_EC2x&EG2x&EG9x&EM05_Series_GNSS_Application_Note
[2] Quectel_UMTS&LTE_EVB_User_Guide
[3] Quectel_EC2x&EG2x&EG9x&EM05_Series_AT_Commands_Manual
[4] Quectel_EC2x&EG2x&EG9x&EM05_Series_QCFG_AT_Commands_Manual
[5] Quectel_EC2x&EG2x&EG9x&EM05_Series_Low_Power_Mode_Application_Note
[6] Quectel_EG9x_Series_AT+QDSIM_Command_Manual
[7] Quectel_RF_Layout_Application_Note
[8] Quectel_EC2x&EG2x&EG9x&EM05_Series_Software_Thermal_Management_Guide
[9] Quectel_Module_Stencil_Design_Requirements
[10] Quectel_Module_SMT_Application_Note

Table 55: Terms and Abbreviations

Abbreviation	Description
3GPP	3rd Generation Partnership Project
ADC	Analog-to-Digital Converter
AGNSS	Assisted GNSS (Global Navigation Satellite System)
AMR	Adaptive Multi-rate
BDS	BeiDou Navigation Satellite System

bps	bits per second
CHAP	Challenge Handshake Authentication Protocol
Cj	Junction Capacitance
CMUX	Connection Multiplexing
CS	Coding Scheme
CSD	Circuit Switched Data
CTS	Clear To Send
DC-HSDPA	Dual-carrier High Speed Downlink Packet Access
DC-HSPA+	Dual-carrier High Speed Packet Access
DCS	Data Coding Scheme
DDR	Double Data Rate
DFOTA	Delta Firmware Upgrade Over-The-Air
DL	Downlink
DRX	Discontinuous Reception
DTR	Data Terminal Ready
DTX	Discontinuous Transmission
EDGE	Enhanced Data Rates for GSM Evolution
EFR	Enhanced Full Rate
EGSM	Enhanced GSM
ESD	Electrostatic Discharge
EVB	Evaluation Board
FDD	Frequency Division Duplex
FILE	File Protocol
FR	Full Rate
FTP	File Transfer Protocol

FTPS	FTP over SSL
GLONASS	Global Navigation Satellite System (Russia)
GMSK	Gaussian Minimum Shift Keying
GNSS	Global Navigation Satellite System
GPRS	General Packet Radio Service
GPS	Global Positioning System
GSM	Global System for Mobile Communications
HR	Half Rate
HSDPA	High Speed Downlink Packet Access
HSPA	High Speed Packet Access
HSUPA	High Speed Uplink Packet Access
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
I/O	Input/Output
Inom	Nominal Current
LED	Light Emitting Diode
LGA	Land Grid Array
LNA	Low Noise Amplifier
LTE	Long Term Evolution
M2M	Machine to Machine
MCS	Modulation and Coding Scheme
MCU	Microcontroller Unit
ME	Mobile Equipment
MIMO	Multiple Input Multiple Output
MMS	Multimedia Messaging Service

MO	Mobile Originated
MQTT	Message Queuing Telemetry Transport
MS	Mobile Station (GSM engine)
MSL	Moisture Sensitivity Level
MT	Mobile Terminated
NAND	Non-volatile Memory Device
NITZ	Network Identity and Time Zone
NMEA	NMEA (National Marine Electronics Association) 0183 Interface Standard
NTP	Network Time Protocol
PA	Power Amplifier
PAM	Pulse-Amplitude Modulation
PAP	Password Authentication Protocol
PCB	Printed Circuit Board
PCM	Pulse Code Modulation
PDU	Protocol Data Unit
PING	Packet Internet Groper
PMIC	Power Management IC
PPP	Point-to-Point Protocol
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
QZSS	Quasi-Zenith Satellite System
RF	Radio Frequency
RHCP	Right Hand Circularly Polarized
RoHS	Restriction of Hazardous Substances
RTS	Request to Send

Rx	Receive
SAW	Surface Acoustic Wave
SMD	Surface Mount Device
SMTP	Simple Mail Transfer Protocol
SMTPS	Simple Mail Transfer Protocol Secure
SMS	Short Message Service
SPI	Serial Peripheral Interface
SSL	Secure Sockets Layer
TCP	Transmission Control Protocol
Tx	Transmit
UART	Universal Asynchronous Receiver/Transmitter
UDP	User Datagram Protocol
UL	Uplink
UMTS	Universal Mobile Telecommunications System
URC	Unsolicited Result Code
(U)SIM	(Universal) Subscriber Identity Module
USB	Universal Serial Bus
Vmax	Maximum Voltage
Vnom	Nominal Voltage
Vmin	Minimum Voltage
V_{IH}max	Maximum High-level Input Voltage
V_{IH}min	Minimum High-level Input Voltage
V_{IL}max	Maximum Low-level Input Voltage
V_{IL}min	Minimum Low-level Input Voltage
$V_I$max	Absolute Maximum Input Voltage

$V_I$min	Absolute Minimum Input Voltage
V_{OH}min	Minimum High-level Output Voltage
V_{OL}max	Maximum Low-level Output Voltage
V_{OL}min	Minimum Low-level Output Voltage
VSWR	Voltage Standing Wave Ratio
WCDMA	Wideband Code Division Multiple Access